Strategic Pricing and the Collapse of Protest Informativeness*

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Abstract

This paper develops a model showing how a monopolist's strategic pricing can endogenously destroy the informational value of consumer protests. Prices act not only as revenue instruments but also as screening devices that determine which consumers remain in the market. Higher signal precision induces the firm to raise prices, excluding low-signal consumers and homogenizing protest incentives. Consequently, protest turnout becomes uninformative precisely when individual signals are more accurate. The mechanism operates under both committed and uncommitted pricing regimes, revealing how market power can undermine collective information aggregation and the disciplinary role of consumer activism.

Keywords: Protest, information aggregation, monopoly pricing

JEL Classification: D42, D72, D81, D82

1. Introduction

Firms increasingly face consumer protests over socially or environmentally controversial products ranging from labor practices and political stances to sustainability concerns.[1] These protests often aim to discipline firms by revealing aggregate consumer sentiment. Yet, protests sometimes fail to convey clear information or to induce

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[1] In the real world, there are many protest activities aiming for product discontinuation, e.g., protests against Nike for labor violations (e.g., "Protesters call on NYU Bookstore to cut ties with Nike," *Washington Square News*, October 13, 2023), a threat against the publication of a book ("Japan firm nixes translation of U.S. book questioning trans surgery," *Japan Times*, December 6, 2023), and so on.

meaningful changes in firm behavior. This paper develops a model to explain one such paradox: when a firm sets prices strategically, consumer protests can become uninformative precisely when consumers are better informed.

The key insight is that prices are not only instruments for extracting revenue but also screening devices that affect which consumers remain in the market. A high price excludes consumers with unfavorable private signals about a product's ethical value. This screening compresses the heterogeneity in protest incentives among those who remain, reducing the informativeness of protest turnout. In equilibrium, the firm's optimal pricing strategy can thus endogenously destroy the information that protests are supposed to reveal.

We study a monopolistic firm selling a product. Consumers receive private signals about the product's underlying value, including moral disutility, and may incur a cost to participate in a protest demanding discontinuation. A noisy public signal of the protest turnout is observed before the firm decides whether to continue selling the product. Crucially, the firm chooses its price either before (committed pricing) or after (uncommitted pricing) observing the public signal. Both decisions feed back into protest incentives.

The model formalizes protest informativeness as the difference in expected turnouts between the cases when the product is valuable and low-value states (including moral disutility). This representation allows clear comparative statics linking price, information precision, and noise in protest turnout. We identify two competing effects of price.

1. A **pivotality effect**: higher prices make the firm's discontinuation threshold more sensitive to turnout, increasing the marginal informational value of participation.
2. An **information-rent effect**: higher prices screen out low-signal consumers, homogenizing pivotal incentives and compressing turnout variation.

We show that when noise is small and private signals are precise, the information-rent effect dominates, and informativeness falls with price. This *Price–Informativeness Trade-off* captures how market structure interacts with collective information transmission.

Specializing in binary signals yields transparent conditions for informative equilibria. We show that informative equilibria exist only at intermediate price levels. When prices are endogenized, a profit-maximizing firm typically selects a high price that eliminates informativeness if consumer signals are sufficiently precise. In both committed and uncommitted pricing regimes, protests thus lose their informational power in precisely those environments where consumers are better informed. This mechanism rationalizes why some firms can withstand or even raise prices amid widespread boycotts or reputational crises.

The framework also highlights welfare and policy implications. Because protests serve as an early-warning mechanism allowing firms to avoid reputational losses, the loss of informativeness under strategic pricing reduces aggregate welfare. Price regulation or commitment devices may therefore restore the social value of protest as an information-transmitting mechanism. These insights bridge industrial organization and political-economy literatures on consumer activism, showing how private market incentives can conflict with social information aggregation.

The rest of the paper is organized as follows: Section 2 presents the model. Section 3 analyzes the equilibrium with continuous signals. Section 4 focuses on a binary signal case and endogenizes the price-setting behavior. Section 5 considers an uncommitted pricing regime. Section 6 discusses the model assumptions and concludes.

1.1. Related Literature

This paper contributes to several strands of literature in industrial organization and political economy.

First, a classic insight in IO is that posted prices are not merely instruments for revenue extraction but also tools to manipulate beliefs and thereby future market behavior. In limit-pricing models, incumbents set prices to influence outsiders' beliefs and actions (e.g., Milgrom and Roberts, 1982, Bagwell and Ramey, 1988). Closely related, in a model of financial markets, Goldstein and Guembel (2008) show a feedback environment where manipulation can reduce the informativeness of prices themselves, weakening their allocational role. Our mechanism differs in that the firm's pricing decision manipulates the informativeness of a separate social signal, protest turnout, by endogenously screening participants and compressing information rent.

Second, we relate to the literature on protest turnout as a signal of latent preferences or information. Lohmann (1993, 1994), Battaglini (2017), and Ekmekci and Lauermann (2022) analyze how collective action can aggregate dispersed information. However, these studies typically consider settings in which individual signals are independent of the market structure, as they consider situations in political contexts. In contrast, we show that in a private market with endogenous pricing, increased signal precision can reduce the informativeness of protest turnout. This arises because the firm uses pricing to screen consumers, which homogenizes participation incentives.

Third, our work contributes to the literature on consumer boycotts and protests. [2] For example, Egorov and Harstad (2017) model a protest as a dynamic war of attrition, while Diermeier and Van Mieghem (2008) and Delacote (2009) study protests as public goods games with fixed thresholds. Our approach differs by emphasizing protests as information transmission mechanisms and considers how pricing behavior endogenously interacts with the informativeness of protests.

Our work adds to the growing literature on protest. Direct evidence for our mechanism is still absent, yet the 2020 boycott of Goya Foods' offers suggestive evidence for our

[2] E.g., Baron (2001), Feddersen and Gilligan (2001), Baron (2003), Innes (2006), Miyagiwa (2009), Glazer et al. (2010), Egorov and Harstad (2017), Peck (2017), and Correa (2025).

mechanism: prices remained unchanged or even rose because the firm could profit from a smaller pool of highly committed buyers, thereby limiting the campaign's impact (see Liaukonyte et al., 2023). The parallel is clear: when the customer base consists mainly of high-WTP consumers, a firm can "price-screen" the market and neutralize the informational power of protest, exactly the channel our model formalizes.

The uncommitted-price case highlights ex-post pricing and the demand-side price-manipulation motive.. When the firm chooses price after observing public signals, consumers may strategically withhold demand (or alter visible participation) to induce markdowns—akin to strategic waiting in dynamic pricing (Su, 2007) and to demand reduction in multi-unit auctions, where buyers shade demand to lower the uniform price (Ausubel et al., 2014). Our price-manipulation motive formalizes a similar logic in a protest environment: participation shifts the public signal in ways that push the firm toward the low-price branch ex post.

2. Model

A firm sells a product to a continuum of consumers with unit demand. The population is normalized to 1. The utility of purchasing the product is $w \cdot v$, where $v \in \{0, 1\}$. We refer to v as *product value* and w as the scale parameter, whose role is discussed later. The value v can be interpreted as incorporating a psychological factor, such as feeling guilty about purchasing a product that causes a problem. We assume that v is unknown to all players. For instance, this uncertainty about v captures situations where a problem of the product has just been revealed, and people are uncertain about how to respond collectively. The prior probability of $v = 1$ is assumed to be $1/2$ for notational simplicity.

Consumers can engage in a protest campaign to stop the sale of the product. Participating in the campaign costs consumer i an amount c_i. The c_i are i.i.d. with distribution function F on $\mathbb{R}$ among consumers (we allow negative costs). We assume that f is the density of F, and $f(0) > 0$. In summary, when the product is sold at price

$w \cdot p$, the consumer's utility is as follows.[3]

$$u_i = [w \cdot (v - p) \cdot \chi(\text{purchase})] \cdot \chi(\text{on sale}) - c_i \cdot \chi(\text{participate in protest}).$$

Before participating in the protest campaign, consumers receive a private signal about the product value v. Consumers receive a private signal represented by the posterior belief $\theta_i \in (0, 1)$, which denotes the conditional probability that $v = 1$. Formally, we define $\theta = \frac{g_1(\theta)}{g_1(\theta)+g_0(\theta)}$, where g_v and G_v denote, respectively, the density and cumulative distribution of the posterior under state v. By this specification, G_v has monotone likelihood ratio property (MLRP): $g_1(\theta)/g_0(\theta)$ is non-decreasing in θ.

Given consumers' private signals, they decide whether to participate in the protest campaign. We assume that consumers act as **ethical agents**. Specifically, they behave according to a **rule-utilitarian** (or **group-utilitarian**) principle, choosing to protest if their participation increases the aggregate welfare of consumers who received the same signal. This framework, which follows Feddersen and Sandroni (2006) and Ekmekci and Lauermann (2022), allows us to directly model the pivotal incentives central to collective action. The precise objective function is given in Section 3.1. A further discussion about the rule-utilitarian assumption is given in Section 6.

One justification for this assumption is that a consumer's participation decision is observed by their social network, leading them to prefer being perceived as ethical. This, in turn, motivates them to act in a way that maximizes consumers' welfare.[4]

However, the adequacy of this justification depends on what participation in the protest entails. On the one hand, if participation involves attending an event organized by activists, the cost may be too high for this motivation to hold. On the other hand, if participation means sharing or liking a post on social media that expresses regret or protest against a firm's product, this motivation seems more plausible. [5]

[3] For each event E, $\chi(E)$ takes the value 1 if E occurs, and 0 otherwise.
[4] Ali and Lin (2013) explores this rationale in the context of voting decisions.
[5] Sharing and liking posts on social media can be enough to compel a firm to discontinue a product. For instance, Adidas withdrew a line of shoes resembling shackles in response to criticism on Facebook ("Adidas cancels 'shackle' shoes after outcry," *CNN*, June 20, 2012).

After the protest, the firm and consumers observe a public signal indicating its scale. Let the actual participation ratio be τ. The firm and consumers observe $t = \tau + \varepsilon$, where ε is a normal noise with 0 mean and σ^2 variance. Let Φ be the standard normal distribution function and φ be its density. ε can be interpreted as noise from media coverage or ambiguity in measuring the protest's true size. For instance, it could reflect noise intentionally created by activists.[6]

Observing the public signal t, the firm decides whether to discontinue the product. We assume that the firm incurs a cost of $wK > 0$ when continuing product sales. This includes distribution and reputation costs associated with selling a product that causes a problem. We also assume that the marginal cost of production is $w\kappa \geq 0$. Therefore, when the product is on sale, the firm's profit is

$$w \cdot [(\text{demand at } p) \cdot (p - \kappa) - K].$$

We normalize the firm's payoffs and write profits without the scale parameter w, as it does not affect the firm's decision. Hence, the firm's profit when the product is on sale is $(\text{demand at } p) \cdot (p - \kappa) - K$.

We assume that $1 > \kappa + K$ to exclude a trivial case. Otherwise, $p = 0$ and not selling the product is optimal.

Finally, the decision flow is as follows. The solution concept is the *perfect Bayesian Nash equilibrium*.

1. The firm sets a price.
2. Each consumer observes a private signal θ_i and decides whether to participate in the protest.
3. Turnout τ is realized, and the firm and consumers observe public signal $t = \tau + \varepsilon$.
4. The firm decides whether to continue selling the product.
5. If the product is on sale, each consumer decides whether to buy it.

[6] This assumption is the same as Ekmekci and Lauermann (2022).

3. Equilibrium with a given price

3.1. Optimal behavior

This section characterizes the optimal behaviors of consumers and the firm after the price is set. We proceed with the analysis as follows: First, consider the consumer's purchase choice, and then, the firm's discontinuation decision. Given these behaviors, we consider the consumer's decision to participate in the protest.

Consumer's purchase decision: We first consider consumers' purchase decisions after public signal observation. Consider the consumer receiving signal θ. The consumer purchases if the expected product value exceeds the price, that is, $\Pr(v = 1 \mid \theta, t) \geq p$.[7]

For notational simplicity, we write $P_\theta(t) = \Pr(v = 1 \mid \theta, t)$. Let λ_v be the true turnout when the realized value is $v \in \{0, 1\}$ and $\lambda = \lambda_0 - \lambda_1$. Then, as $t = \lambda_v + \varepsilon$ at state v, $P_\theta(t)$ is written as follows (recall that φ is the pdf of the standard normal distribution).

$$P_\theta(t) = \frac{\varphi\left(\frac{t-\lambda_1}{\sigma}\right)\theta}{\varphi\left(\frac{t-\lambda_1}{\sigma}\right)\theta + \varphi\left(\frac{t-\lambda_0}{\sigma}\right)(1-\theta)}.$$

By using this formula, when we assume $\lambda_0 > \lambda_1$, inequality $P_\theta(t) > p$ is rewritten as

$$t < \frac{\lambda_0 + \lambda_1}{2} + \frac{\sigma^2}{2\lambda}\left[\ln\left(\frac{\theta}{1-\theta}\right) - \ln\left(\frac{p}{1-p}\right)\right] =: T_\theta(p).$$

In other words, $T_\theta(p)$ is the supremum value of public signal t with which the type-θ consumers purchase the product. The following lemma shows that high-θ consumers are more likely to purchase the product than low-θ consumers.

Lemma 1. *If $\lambda_0 > \lambda_1$, $T_\theta(p)$ is increasing in θ for any $p \in (0, 1)$.*

By this lemma, $T_\theta(p)$ determines another cutoff: Let $\Theta_t(p)$ be the minimum θ such that $T_\theta(p) > t$. Then,

[7] Note that as this decision is in the last stage of the game, the purchase decision does not affect others' utilities. Thus, each consumer's purchase decision is based only on their utility, although the consumers have welfare concerns.

$$\Theta_t(p) = \psi\left(\frac{2\lambda}{\sigma^2}\left[t - \frac{\lambda_0 + \lambda_1}{2}\right] + \ln\left(\frac{p}{1-p}\right)\right),$$

where $\psi(x) = \exp(x)/(1 + \exp(x))$. Therefore, when the product value is v, the demand for the product is $1 - G_v(\Theta_t(p))$.

The firm's discontinuation decision: Given the consumer's choice after observing public signal t, this section considers the condition under which the firm discontinues the product sale. We focus on a cutoff equilibrium; there is a cutoff $T_f(p)$ such that the firm discontinues the product sale if and only if $t > T_f(p)$. To determine $T_f(p)$, let $\Pi(p, t)$ be the expected net-profit of the firm after observing public signal t:

$$\Pi(p, t) = \sum_{v \in \{0,1\}} \Pr(v \mid t)[1 - G_v(\Theta_t(p))](p - \kappa) - K.$$

Next, we show that the cutoff can be decomposed into (λ, t, σ)-dependent and (λ, t, σ)-independent parts. Let $a = \Pr(v = 0 \mid t)/\Pr(v = 1 \mid t) = \frac{\varphi((t-\lambda_0)/\sigma)}{\varphi((t-\lambda_1)/\sigma)} = \exp\big(\lambda\sigma^{-2}[t - (\lambda_0 + \lambda_1)/2]\big)$ be the likelihood ratio of the product value conditioned on public signal t. Then, the profit is rewritten as

$$\Pi(p, t) = \Pi^*(p, a) := \frac{p - \kappa}{1 + a}\big[\big(1 - \tilde{G}_1(\omega(p, a))\big) + a\big(1 - \tilde{G}_0(\omega(p, a))\big)\big] - K,$$

where $\omega(p, a) = 2\ln(a) + \ln(p/(1-p))$ and $\tilde{G}_v = G_v \circ \psi$. In other words, a is a sufficient statistic for determining the profit. Let $a^*(p)$ be the solution of $\Pi^*(p, a) = 0$ with respect to a, which is independent of λ, t, and σ. Then, the firm's cutoff is written as follows: $T_f(p) = \sigma^2\lambda^{-1}\ln(a^*(p)) + (\lambda_0 + \lambda_1)/2$.

Consumer's participation decision: Before the public signal observation, the consumer chooses whether to participate in the protest. This section examines that condition and how informative turnout is about the product value.

We focus on the cutoff strategy: each type-θ consumer participates in the protest if and only if the marginal benefit of participation exceeds its cost. As mentioned in the model section, we consider agents that follow rule-utilitarian: they maximize the

aggregate utility of the same type of consumers written in the following formula.[8, 9]

$$\int^{T_f(p)-\mathrm{d}\tau} w \cdot [P_\theta(t) - p]_+ \cdot \xi_\theta(t)\,\mathrm{d}t - c_i\,\mathrm{d}\tau - \sum_{j \text{ is a participant}} c_j,$$

where we define $[\cdot]_+ = \max\{\cdot, 0\}$ and $\xi_\theta(t) = \sigma^{-1} \cdot [\theta \cdot \varphi((t-\lambda_1)/\sigma) + (1-\theta) \cdot \varphi((t-\lambda_0)/\sigma)]$. The term $\mathrm{d}\tau$ represents a marginal increase in the participation rate. In this formula, the firm applies a cutoff criterion: discontinuing the product if and only if $t > T_f(p)$. A consumer's participation increases turnout by $\mathrm{d}\tau$, which also increases their costs by $c_i\,\mathrm{d}\tau$. Then, consumer i participates in the protest if and only if

$$\underbrace{-w \cdot \Big[P_\theta\big(T_f(p)\big) - p\Big]_+ \cdot \xi_\theta\big(T_f(p)\big)}_{\text{marginal benefit of participation}} \geq c_i.$$

Now let $c_\theta(p) = -w\Big[P_\theta\big(T_f(p)\big) - p\Big]_+ \cdot \xi_\theta\big(T_f(p)\big)$ be the cutoff of type-θ consumer. The next subsection characterizes the equilibrium and performs comparative statics.

3.2. Equilibrium and protest informativeness

Given the optimal behavior characterized above, this subsection studies the continuation-game equilibrium at a given price p. Then, we define and analyze the informativeness of the protest.

In the previous subsection, to determine the optimal behavior, the value $\lambda = \lambda_0 - \lambda_1$ was exogenously given. We endogenize this value. As the cutoff of participation decision is $c_\theta(p)$, λ_v, the participation rate when the product value is v is calculated as

$$\lambda_v = \int F(c_\theta(p)) g_v(\theta)\,\mathrm{d}\theta.$$

As $c_\theta(p)$ depends on λ, define $\Lambda(\lambda, p)$ as follows:

$$\Lambda(\lambda, p) = \lambda_0 - \lambda_1 = \int F(c_\theta(p))(g_0(\theta) - g_1(\theta))\,\mathrm{d}\theta.$$

[8] Note that assuming this utility does not affect the purchasing decision, as it does not affect the other consumers' utility and the firm's decision.

[9] Our results continue to hold even when each consumer maximizes overall consumer welfare. See Section 6.

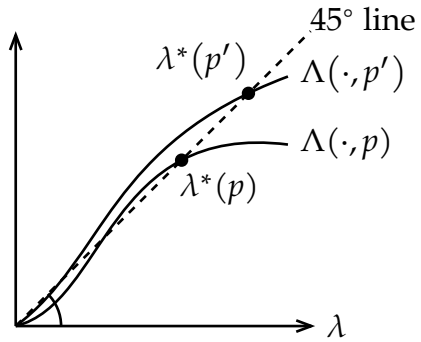

Figure 1: $\lambda^*(p)$ indicates the largest fixed point of $\Lambda(\cdot, p)$

The equilibrium under price p is characterized as a fixed point of $\Lambda(\cdot, p)$, that is, $\lambda^*(p) = \Lambda(\lambda^*(p), p)$.

We refer to λ^* as the **informativeness** of the protest.[10] Indeed, we have[11]

Fact 1. *The distribution of ex post belief $\{P_\theta(t)\}_{t\in\mathbb{R}}$ with higher λ is a mean-preserving spread of that with lower λ.*

We say that the protest equilibrium is *uninformative* if $\lambda^*(p) = 0$ and *informative* if $\lambda^*(p) > 0$.

It is important to note that the uninformative equilibrium exists for any parameter values. If the firm sets $T = \infty$ or $T = -\infty$, the protest does not affect whether the product sale continues. This leads to both types of consumers having indifferent incentives. Conversely, if the protest is uninformative, the firm's decision remains unchanged regardless of the turnout. This implies that $T = \infty$ or $T = -\infty$.

Therefore, we focus on the most informative equilibria to obtain clear comparative statics. Especially, to focus on the most informative equilibrium, we consider the largest fixed point of $\Lambda(\cdot, p)$.

As we focus on the largest fixed point, if $\Lambda(\lambda, p)$ is (weakly) increasing in p pointwise, then $\lambda^*(p)$ is non-decreasing in p; likewise, if $\Lambda(\lambda, p)$ is (weakly) decreasing in p, then $\lambda^*(p)$ is non-increasing in p (see Figure 1 for an illustration).[12]

[10] This definition is the same as Ekmekci and Lauermann (2022).
[11] Although this fact is standard, we provide a proof in Appendix B for completeness.
[12] This fact is standard. Furthermore, this fixed point is stable under adaptive dynamics.

The following theorem provides a sufficient condition that determines the sign of $\partial\Lambda(\lambda, p)/\partial p$.

Theorem 1 (Price-Informativeness Trade-off). *Let* $\eta_a(p) := \frac{\partial a^*(p)/\partial p}{a^*(p)}$. $\partial\Lambda(\lambda, p)/\partial p < 0$ *for sufficiently small* σ *if*

$$(1 - a^*(p))\left[1 + \left[\frac{p - \frac{1}{2}}{a^*(p) - 1} + \frac{p}{2}\right]\eta_a(p)\right] > 0.$$

To capture the intuition for this result, recall that the cutoff is written as follows:

$$c_\theta(p) = -w\underbrace{\Big[P_\theta\big(T_f(p)\big) - p\Big]_+}_{\text{Information rent}} \cdot \underbrace{\xi_\theta\big(T_f(p)\big)}_{\text{Pivotal prob.}}.$$

The price affects in two ways: one is on the *information rent*, and the other is on the probability of being *pivotal*.

The pivotal effect is determined by how the price affects the firm's cutoff $T_f(p)$, whose effect can be either positive or negative.

In contrast, the effect on information rent is $w[1 - P_\theta(1 - P_\theta)\eta_a(p)]$, which decreases with $\eta_a(p)$, the elasticity of a^*. The sufficient condition given in Theorem 1 specifies when the information-rent effect dominates the pivotal effect. In this case, λ decreases because the difference in c_θ is compressed.[13]

Finally, we consider the optimal price that maximizes the total profit. Fix λ and define

$$\Pi_\lambda(p) = \frac{1}{2\sigma}\int_{-\infty}^{T_f(p)} \Pi(p, t)\Big[\varphi\Big(\tfrac{t-\lambda_1}{\sigma}\Big) + \varphi\Big(\tfrac{t-\lambda_0}{\sigma}\Big)\Big]\,\mathrm{d}t.$$

Then, $p^*_\sigma = \arg\max \Pi_\lambda(p)$ is the optimal price when the effect of λ is ignored.

Lemma 2. *Suppose that* λ *is given. Let* $p_a = \arg\max_p \Pi^*(p, a)$. *Then, when* $\sigma \to 0$ *or* $\sigma \to \infty$, p^*_σ *converges to* $p_{a^\dagger_\sigma}$ *where* $a^\dagger_\sigma = \exp(\lambda/\sigma)$.

[13] This result depends on the low-noise environments. In contrast to this result, when σ is large, the information-rent effect becomes negligible, and the pivotal effect dominates. As a result, the informativeness increases with the price. See Appendix C for details.

To shorten notation, $p_\sigma^\dagger := p_{a_\sigma^\dagger} = \arg\max_p \Pi^*(p, a_\sigma^\dagger)$. By this lemma, to know $\partial\Lambda/\partial p$ in a high or low noise environment, we only need to consider $a^*(p_\sigma^\dagger)$ and apply Theorem 1.

In a low-noise environment (i.e., $\sigma \to 0$), we show that $p_{a_\sigma^\dagger} \to 1$. Intuitively, when noise is small, the protest almost perfectly reveals the realized value v. If $v = 1$, setting $p \approx 1$ is optimal. If $v = 0$, not selling the product is optimal. In either case, setting $p \approx 1$ is approximately dominant. Under some conditions, we show that price and protest informativeness conflict: the optimal price should be lower when the firm prefers more informative protests.

Proposition 1. *Suppose that* $\tilde{g}_v(x) > 0$ *and* $\lim_{x\to\infty}\left(1 - \tilde{G}_v(x)\right)/\tilde{g}_v(x) = \infty$.[14] *Then,* $\lim_{\sigma\to 0} \partial\Lambda(\lambda, p)/\partial p|_{p=p_\sigma^\dagger} < 0$.

4. Full-equilibrium characterization with binary signals

From this section, we assume a binary signal to characterize the whole equilibrium. That is, G_v has a Dirac mass on $\{\theta_H, \theta_L\}$. Specifically, we assume that $P(\theta_i = \theta_H \mid v = 1) = P(\theta_i = \theta_L \mid v = 0) = \mu \in (1/2, 1)$. We refer to μ as *signal precision*. Then, the posteriors are $\theta_H = \mu$ and $\theta_L = 1 - \mu$.

4.1. Equilibrium characterization

By the binary-signal simplification, we provide the explicit form of $T_f(p)$. Hereafter, to simplify the notation, $T_{\theta_H} = T_H$, $c_{\theta_H} = c_H$, and we refer to a type-θ_H consumer as a type-H consumer. The same simplification is applied to θ_L. Now the following lemma characterizes the firm's cutoff.

Lemma 3. *Suppose that* $T_L(p) < T_H(p)$, *and let*

$$\tilde{T} := \frac{\lambda_0 + \lambda_1}{2} + \frac{\sigma^2}{2}\frac{\ln\left(\frac{\mu - \frac{K}{p-\kappa}}{\frac{K}{p-\kappa} - (1-\mu)}\right)}{\lambda_0 - \lambda_1}.$$

Then, the cutoffs are classified as follows.

[14] The conditions $\tilde{g}_v > 0$ assumes the full support of posterior distribution, and $\lim_{x\to\infty}\left(1 - \tilde{G}_v(x)\right)/\tilde{g}_v(x)$ assumes that signal distributions have a heavy tail: equivalently, the hazard rate vanishes as $x \to \infty$.

(i) *If* $p - \kappa \geq K/(1-\mu)$, $T_f(p) = T_H(p)$.

(ii) *If* $K/(1-\mu) > p - \kappa > K/\mu$, $T_f(p) = \max\{\min\{T_H(p), \tilde{T}\}, T_L(p)\}$.

(iii) *If* $K/\mu \geq p - \kappa \geq K$, $T_f(p) = T_L(p)$.

(iv) *If* $K > p - \kappa$, $T_f(p) = -\infty$.

Our discussion is guided by the following proposition, which characterizes the range of prices for which an informative equilibrium exists: An informative equilibrium exists only at intermediate prices, showing a clear price–informativeness trade-off.. This proposition is a key tool in our analysis.

Proposition 2. *Let* $\overline{p} := \mu(K + (1-\mu)\kappa)/(\mu(1-\mu) + (2\mu - 1)K)$. *(a) If* $p \geq \overline{p}$ *or* $p \leq K + \kappa$, *no informative equilibrium exists. (b) Suppose that* $K + \kappa < p < \overline{p}$, $f(c) > 0$ *for any* $c \in \mathbb{R}$, *and* $F(0) > 0$. *Then, if* w *is sufficiently large, an informative equilibrium exists. Moreover the equilibrium* $\lambda^* \geq (2\mu - 1)F(0) - \delta$ *for arbitrary* $\delta > 0$ *by taking large* w.

An important consequence of Proposition 2 is that it reveals a fundamental commitment problem faced by the firm: for the equilibrium to be informative, the price should be below $\overline{p}$. This price ceiling reveals a crucial tension. Even if a protest could perfectly reveal the state (e.g., $v = 1$), the firm cannot fully exploit this information by setting a price close to 1. The reason is that the protest's informativeness is endogenous: the very act of setting a high price would, by the logic of Proposition 2, render the protest uninformative in the first place. This dynamic has a perverse consequence for consumers. Note that the price ceiling for an informative protest, $\overline{p}$, increases with signal precision μ if, for example, $\mu > \sqrt{\kappa + K}$.[15] In such a case, more precise private information does not necessarily benefit consumers; it may simply empower the firm to maintain a higher price.

[15] Indeed, by a differentiation, $\overline{p}$ is increasing in μ if and only if $-2\kappa\mu(1-\mu) - \mu^2 + \kappa + K < 0$.

4.2. Firm's optimal pricing

Assuming that for any given price, the most informative equilibrium prevails, this section endogenizes the firm's price choice to determine the overall equilibrium outcome.

First, as a benchmark, we consider the model without protest. In the benchmark case, the type-H's expected product value is $E[v] = \mu$, and that of the type-L consumers is $E[v] = 1 - \mu$. If the firm keeps the product sale, the optimal price is $p = \mu$ or $p = 1 - \mu$. The expected profit when $p = \mu$ is $(\mu - \kappa)/2 - K$, and that when $p = 1 - \mu$ is $1 - \mu - \kappa - K$.

Now, we consider the firm's optimal pricing with the protest. The following proposition shows that the optimal price is $p = \mu$ when μ is high enough, in which case the protest is uninformative.

Proposition 3. *Suppose that $\mu > \kappa + 4K$, $\kappa + K > 1/2$ and $M(\mu) > 0$, where*

$$M(\mu) := \mu - \kappa - K\frac{\mu + \kappa(2\mu - 1)}{\mu(1 - \mu) + (2\mu - 1)K} - K.$$

Then, $p = \mu$ is optimal and the resulting equilibrium is uninformative.

As illustrated in Figure 2, $M(\mu) > 0$ is possible for intermediate μ when K is small.

The intuition for this result hinges on the firm's trade-off between information acquisition and surplus extraction. As Proposition 2 shows, informative equilibrium requires the price to be $\overline{p}$. The condition $M(\mu) > 0$ ensures that $\overline{p}$ is small enough, in which case, setting $p = \mu$ and an uninformative equilibrium yields a greater profit. The information is valuable for the firm when K is high. Therefore, the uninformative equilibrium with a high price is preferable when K is low.

The above observation depends on whether an uninformative equilibrium with $p = \mu$ yields a sufficient profit. If not, an informative equilibrium leads to a higher profit, as the following proposition shows.

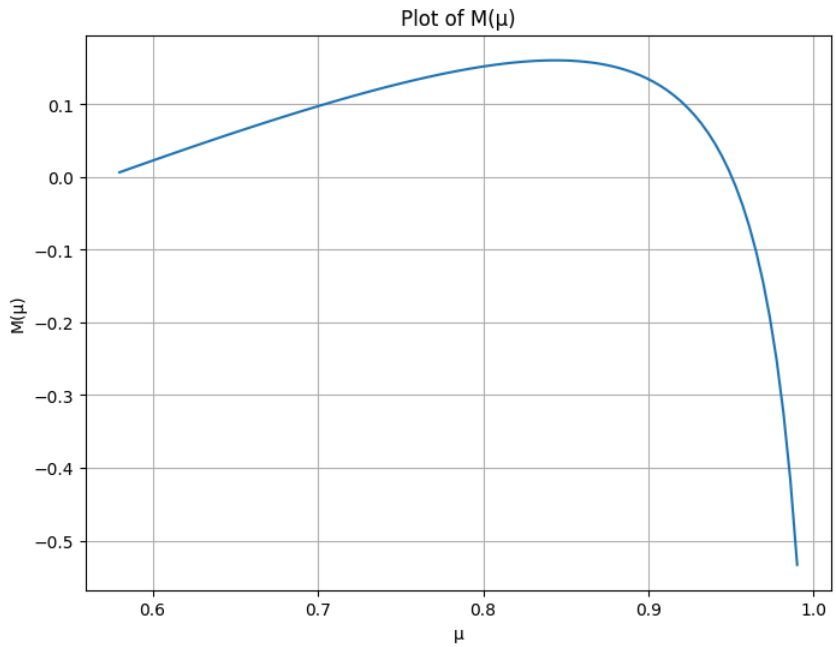

Figure 2: Graph of $M(\mu)$ when $\kappa = 1/2$, and $K = 0.02$

Proposition 4. *Suppose that $\mu \in (K + \kappa, 2K + \kappa)$. Then, for sufficiently small σ, there is $\overline{w}$ such that for each $w > \overline{w}$, the protest is informative at the firm's optimal price.*

With the condition of Proposition 4, $\overline{p} > \mu$, and the uninformative equilibrium yields a deficit. Therefore, setting the price to $p = \overline{p}$ induces an informative equilibrium that yields a positive profit.

By Propositions 3 and 4, we observe that the resulting equilibrium is uninformative for a higher μ, while it is informative for a lower μ. Theorem 2 summarizes the result.

Theorem 2. *Suppose that $\kappa + K > 1/2$. For $\mu \in (K + \kappa, 2K + \kappa)$, an informative equilibrium exists for sufficiently large w and small σ. For $\mu > 4K + \kappa$ with $M(\mu) > 0$, no informative equilibrium exists.*

Remark 1 (Blackwell paradox under endogenous pricing). It is worth noting that Theorem 2 is not an artifact of assuming binary signals. Let G_B be the posterior distribution under the binary signal with signal precision $\mu > 4K + \kappa$ with $M(\mu) > 0$ used in Theorem 2, and G_C be a continuous posterior distribution with support $[1 - \mu, \mu]$ that is Blackwell-dominated by G_B. Then, although G_B is strictly more informative

than G_C in the Blackwell sense, the equilibrium protest under endogenous pricing can be less informative ($\lambda^*_G(p^*(G))$ denotes the highest equilibrium informativeness under signal distribution G):

$$0 = \lambda^*_{G_B}(p^*(G_B)) \leq \lambda^*_{G_C}(p^*(G_C)).$$

In other words, a more informative individual-signal structure can yield a less informative collective outcome once the firm optimally chooses its price.

This contrasts sharply with a model of protest against the policy-setting government. Ekmekci and Lauermann (2022) show that each person's signal informativeness positively correlates with the protest campaign's informativeness.

4.3. Welfare Analysis

This discussion examines social welfare, which is the sum of the firm's profit and consumer surplus. Then, social welfare is $((v - \kappa) \cdot \text{purchase} - K)\chi(\text{on sale})$. In an informative equilibrium with sufficiently small σ, the protest approximately perfectly reveals the true state v regardless of the size of μ. Therefore, social welfare is $(1 - \kappa - K)/2$. On the contrary, in an uninformative equilibrium, social welfare is at most $(\mu - \kappa)/2 - K$. Therefore, the informative equilibrium attains a higher social welfare.

Our findings suggest that as the consumers' private signals become precise, social welfare can decrease as the informativeness λ vanishes. The following corollary is a summary.

Corollary 1. *Suppose that $\kappa + K > 1/2$. When σ is sufficiently small and w is large enough, social welfare is larger for $\mu \in (K + \kappa, 2K + \kappa)$ than that for $\mu > 4K + \kappa$ with $M(\mu) > 0$.*

5. Uncommitted price

In the previous section, the firm committed to a price p before observing the public signal; here, we allow it to set the price ex post. Therefore, the new decision flow is as follows:

1. Each consumer observes a private signal θ_i and decides whether to participate in the protest.
2. Turnout τ realizes, and the firm and consumers observe public signal $t = \tau + \varepsilon$.
3. The firm decides whether to continue selling the product, and if so, sets a price.
4. If the product is on sale, each consumer decides whether to buy it.

A crucial difference from the committed-price case is that turnout now influences the price: informative protests push the firm toward the low-price branch.[16] Then, protest participation has two effects on type-H consumers' utility. On one hand, a protest threatens to eliminate their potential informational rents, discouraging participation. On the other hand, a larger protest can signal a lower product value to the firm, inducing it to set a lower price. This "price-manipulation" motive encourages type-H participation. The overall effect on equilibrium informativeness depends on the relative strength of these competing incentives.

We now examine whether the earlier conclusion still holds; the previous section shows that with higher μ, the equilibrium is less likely to be informative. We show that even in the uncommitted price case, a similar conclusion is drawn. The next subsection provides the formal analysis.

5.1. Formal analysis of the uncommitted price case

To focus on the discussion on whether an informative equilibrium exists, we assume that $\lambda_0 > \lambda_1$. Then we verify the consumers' incentives to participate in the protests. Note that even in the uncommitted price case, after the price is set, the consumers' decision criterion for purchasing the product is unchanged. Thus, type-θ consumer purchases the product if and only if $t < T_\theta(p)$.

Given this strategy, the firm's profit is

[16] Indeed, protests tend to decrease prices. Hendel et al. (2017) empirically demonstrated that price declines after a consumer boycott, organized in Israel on Facebook.

$$\Pi(p,t) = \begin{cases} -K & \text{if } t > T_H(p) \\ [\Pr(v=1 \mid t)\cdot\mu + \Pr(v=0 \mid t)\cdot(1-\mu)]\cdot(p-\kappa) - K & \text{if } t \in (T_L(p), T_H(p)] \\ p-\kappa-K & \text{if } t \le T_L(p). \end{cases}$$

This implies that the optimal price is either $p = P_H(t)$ or $P_L(t)$. Therefore, we need to compare the following equations.

$$\Pi_H^*(t) := [\Pr(v=1 \mid t)\cdot(2\mu-1)+1-\mu)]\cdot(P_H(t)-\kappa) - K$$

$$\Pi_L^*(t) := P_L(t) - \kappa - K.$$

$\Pi_H^*(t)$ represents the profit when the firm focuses solely on type-H consumers, while $\Pi_L^*(t)$ indicates the profit when targeting both consumer types. Also, the firm discontinues the product sales rather than selling it when $\max\{\Pi_H^*, \Pi_L^*\} < 0$.

To simplify the discussion, we assume $\kappa = 0$ hereafter. The following lemma identifies the condition under which one profit exceeds the other.

Lemma 4. *Assume that $\kappa = 0$. Then, there exist values μ^* and $\mu^\dagger$ with $\mu^* < \mu^\dagger$ such that: (a) If $\mu > \mu^\dagger$, there is $\hat{t} < T_L(K)$ such that $\Pi_H^*(t) < \Pi_L^*(t)$ if and only if $t < \hat{t}$. (b) If $\mu < \mu^\dagger$, $\Pi_H^*(t) < \Pi_L^*(t)$ for each $t < T_L(K)$. (c) $T_L(K) \to \infty$ if $\mu < \mu^*$, and $T_L(K) \to -\infty$ if $\mu > \mu^*$.*

In words, Lemma 4 shows that when μ is high, the firm sets $p = P_L$ for low t, $p = P_H$ for medium t, and discontinues for high t. If μ is intermediate, $p = P_L$ is optimal for low t and discontinuation is optimal for high t.

Given the firm's behavior, we next consider consumers' incentives to participate in the protest. When μ is high, type-H consumers still buy near the firm's cutoff, so they have stronger incentives for price manipulation than type-L consumers, reversing the participation pattern and rendering protests uninformative. When μ is low, both types purchase while the product is sold, preserving $c_L > c_H$ and keeping protests informative. The next theorem formalizes these two regimes of μ.

Theorem 3. *Assume that $c_L > c_H$.*

(a) Suppose that $\mu > \max\{\mu^\dagger, 2K\}$. Then, for sufficiently large σ, $c_H > c_L$ and for sufficiently small σ, $c_L > c_H$.

(b) Suppose that $\mu \in (\mu^*, \mu^\dagger)$. Then, for sufficiently large σ, $c_L > c_H$.

Note that inequality $c_H > c_L$ is gained by the supposition that $\lambda_0 > \lambda_1$, which means that $c_L > c_H$. This is a contradiction. So, the equilibrium should be uninformative.

Hence, protests become uninformative when μ is high and remain informative for intermediate μ for large w, mirroring the pattern under committed pricing.

Lastly, note that the price-manipulation incentive disappears when the noise level σ is small. In this case, even a marginal increase in participation sends a sufficiently strong signal that $v = 0$, prompting the firm to discontinue the product immediately and thus eliminating any incentive to manipulate prices.

6. Conclusion and Discussions

This paper identifies the conditions under which price and protest informativeness conflict and characterizes the circumstances under which protest campaigns successfully aggregate dispersed consumer information.

The effect of price on protest informativeness can be decomposed into two mechanisms: the pivotal effect and the information rent effect. In low-noise environments, the information-rent effect dominates, creating a conflict between price and informativeness.

By focusing on a binary-signal framework, we show that higher precision in consumers' private signals does not necessarily enhance the informativeness of protests. Paradoxically, consumers' limited information—or even their ignorance—can improve collective information aggregation by preventing excessive price-based screening.

Our results yield clear policy implications. Laissez-faire pricing can endogenously destroy the informational value of consumer protests. Informative protest equilibria exist only within an intermediate price window. When signal precision is high, profit-maximizing firms set prices that induce uninformative outcomes. Because the informative

equilibrium yields higher social welfare, modest price-cap or price-commitment policies that keep prices inside the informative window can improve welfare by preserving the disciplinary and informational role of consumer activism.

Importantly, this mechanism also operates when prices are set ex post. Under uncommitted pricing, protests become uninformative in high-noise environments because a price-manipulation incentive dominates. However, when public signals of protest turnout are precise, by reflecting transparent or verified measurement, the pivotal-information effect prevails, which restores informativeness. Hence, improving the accuracy and transparency of protest reporting can serve as a policy substitute for direct price regulation, ensuring that consumer activism continues to function as an effective information-transmitting institution.

In the remainder of this study, we discuss the assumptions in our model.

Objective of participation in protest: We assumed that consumers consider their type's aggregate utility when deciding whether to participate in the protest. However, as in Ali and Lin (2013), if this motivation stems from a reputation concern, it is more plausible to assume that consumers consider the expected overall consumer welfare.

Consider a scenario where each consumer's decision to participate in the protest is based on maximizing the expected consumer welfare. The consumer's belief, represented by $\theta \in \{\theta_H, \theta_L\}$, influences their expected consumer welfare as follows.

$$\int^{T_f(p)-\mathrm{d}\tau} w\Big[\theta \cdot [P_\theta(t) - p]_+ \xi_\theta(t) + (1-\theta) \cdot [P_{\theta'}(t) - p]_+ \xi_{\theta'}(t)\Big]\,\mathrm{d}t,$$

where $\theta' \in \{\theta_H, \theta_L\}$ with $\theta' \neq \theta$. In this case, the type-H participates in the protest if and only if $c_i \leq c_H^* := \mu c_H + (1-\mu)c_L$ while the type-L does if and only if $c_i \leq c_L^* := (1-\mu)c_H + \mu c_L$. As long as $c_L > c_H$, the type-L consumers are more likely to participate. We can apply a similar discussion to obtain the same conclusions.

A potential concern is our assumption that consumers act as 'ethical agents' when deciding whether to protest. We argue that this assumption, while a simplification,

allows for a clear analysis of our main contribution without loss of generality for the core economic trade-off.

First, this framework is a standard and convenient method for directly modeling pivotal incentives, which are central to any model of collective action. The main driver of our results is not the ethical motivation itself, but how the firm's pricing strategy alters the consumers' expected utility of being pivotal. By homogenizing the incentives of high-signal and low-signal consumers at the firm's decision threshold, the firm's pricing, not the specific form of consumer motivation, renders the protest uninformative.

Second, we conjecture that our results are robust and would hold in a setting with purely self-interested consumers. Following the approach in Battaglini (2017) and Ekmekci and Lauermann (2022), one could model a finite n-player game where each agent strategically calculates their probability of being pivotal. However, the key distinction and contribution of our paper is the introduction of a strategic price-setting firm. In such a model, consumers' calculations would have to incorporate the firm's pricing rule, which depends on the informativeness of the protest. The central mechanism we identify—that high signal precision incentivizes the firm to set a high price, which in turn screens the market and dampens information aggregation—would therefore remain. Consequently, the fundamental paradox highlighted in this paper is expected to persist.

Modeling of a protest: Our model focuses on protest as an expressive act, distinct from the purchase decision itself (a boycott). This focus is motivated by the many contemporary consumer campaigns where public expression on platforms like social media is a central feature. Our contribution is to show how information aggregation can fail at this expressive stage. A richer model could integrate both channels—expression and purchasing—to study their dynamic interplay.

Mechanism design by the firm: In our model, the firm's pricing strategy is limited to setting a price. While a firm can indeed use its price-posting mechanism to elicit information, our analysis reveals that the two mechanisms of information —protest and pricing— are in direct conflict. The firm's profit-maximizing use of its pricing power to screen consumers is precisely what destroys the informational value of the protest. This tension between the private incentive to extract surplus and the collective goal of information aggregation lies at the heart of our findings and remains a fruitful area for further exploration.

Furthermore, there is a fundamental limitation to using a price-posting mechanism for information elicitation related to the timing of information. In our model, the firm must incur a cost K to continue selling the product. Therefore, any strategy to learn from realized demand requires the firm to take a risk for this cost. If the product value turns out to be low ($v = 0$), the information arrives 'too late,' only after the cost K has been sunk.

In contrast, the protest mechanism provides a signal before this irreversible cost is paid. An informative protest, therefore, serves as a valuable, ex-ante signal that allows the firm to avoid incurring a loss, a benefit that a post-entry 'price experiment' cannot offer. This highlights the unique economic importance of protests as a pre-commitment signaling device, justifying our focus on their information aggregation properties.

Appendix

A. Proofs

A.1. Proofs in Section 3

Proof of Theorem 1. We first show the following claim, which summarizes the limits of pivotal probability, ξ_θ.

Claim 1.

$$\lim_{\sigma\to 0}\xi_\theta\big(T_f(p)\big)=0 \text{ and } \lim_{\sigma\to 0}\frac{\xi'_\theta\big(T_f(p)\big)}{\xi_\theta\big(T_f(p)\big)}T'_f(p)=-\frac{1}{2}\frac{\theta-(1-\theta)a^*(p)}{\theta+(1-\theta)a^*(p)}\eta_a(p).$$

Proof of Claim 1. The first limit follows from equations $\big(T_f(p)-\lambda_1\big)/\sigma=\lambda(2\sigma)^{-1}+\sigma\lambda^{-1}\ln(a^*(p))$, and $\xi_\theta\big(T_f(p)\big)=\sigma^{-1}\varphi\big(\big(T_f(p)-\lambda_1\big)/\sigma\big)[\theta+(1-\theta)a]$, which implies $\lim_{\sigma\to 0}\xi_\theta\big(T_f(p)\big)=0$.

The second limit follows from equations $T'_f(p)=\sigma^2\lambda^{-1}\eta_a(p)$ and

$$\frac{\xi'_\theta\big(T_f(p)\big)}{\xi_\theta\big(T_f(p)\big)}=-\frac{\theta\frac{T_f(p)-\lambda_1}{\sigma^2}+(1-\theta)\frac{T_f(p)-\lambda_0}{\sigma^2}a}{\theta+(1-\theta)a}.$$

Since $T_f(p)-\lambda_1\to\lambda/2$ and $T_f(p)-\lambda_0\to-\lambda/2$ as $\sigma\to 0$, we conclude the second limit. ■

The next claim shows convergence rates of the derivatives of c_θ.

Claim 2. *(1) $|\partial c_\theta(p)/\partial p|\propto\left|\xi_\theta\big(T_f(p)\big)\right|$ if $\theta=\theta(p)$, and (2) for each $\theta>\theta(p)$, (2-1) $|\partial c_\theta(p)/\partial p|\propto\left|\xi'_\theta\big(T_f(p)\big)T'_f(p)\right|$, (2-2) $|\partial c_\theta(p)/\partial\theta|\propto\left|\xi_\theta\big(T_f(p)\big)\right|$, and (2-3) $\left|\partial^2 c_\theta(p)/\partial\theta\partial p\right|\propto\left|\xi'_\theta\big(T_f(p)\big)T'_f(p)\right|$.*

Proof of Claim 2. We have two cases. First, suppose that $P_\theta\big(T_f(p)\big)>p$. Then, we have

$$\frac{\partial c_\theta(p)}{\partial p}=\xi_\theta\big(T_f(p)\big)w\big(1-P'_\theta\big(T_f(p)\big)T'_f(p)\big)-w\big[P_\theta\big(T_f(p)\big)-p\big]_+\xi'_\theta\big(T_f(p)\big)T'_f(p)$$

$$P'_\theta\big(T_f(p)\big)T'_f(p)=-P_\theta(1-P_\theta)\eta_a(p)<\infty$$

Then, $|\partial c_\theta(p)/\partial p|\propto\left|\xi'_\theta\big(T_f(p)\big)T'_f(p)\right|$ if $\theta>\theta(p)$ and $\lim_{\theta\nearrow\theta(p)}|\partial c_\theta(p)/\partial p|\propto\left|\xi_\theta\big(T_f(p)\big)\right|$.

Next, $\partial c_\theta/\partial\theta$ is calculated as

$$\frac{\partial c_\theta}{\partial \theta} = -\xi_\theta\big(T_f(p)\big)w\left[\frac{\theta}{[\theta+(1-\theta)a^*(p)]^2} - \big[P_\theta\big(T_f(p)\big)-p\big]_+ \frac{\varphi\Big(\frac{T_f(p)-\lambda_1}{\sigma}\Big)-\varphi\Big(\frac{T_f(p)-\lambda_0}{\sigma}\Big)}{\sigma\xi_\theta\big(T_f(p)\big)}\right].$$

Note that

$$\frac{1}{\sigma}\frac{\varphi\Big(\frac{T_f(p)-\lambda_1}{\sigma}\Big)-\varphi\Big(\frac{T_f(p)-\lambda_0}{\sigma}\Big)}{\xi_\theta\big(T_f(p)\big)} = \frac{1-a^*(p)}{\theta+(1-\theta)a^*(p)},$$

which is independent of σ. Therefore, $|\partial c_\theta/\partial\theta| \propto \xi_\theta\big(T_f(p)\big)$.

Finally, the cross derivative is calculated as

$$\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} = -\xi_\theta'\big(T_f(p)\big)T_f'(p)Y(p) - \xi_\theta\big(T_f(p)\big)Y'(p),$$

$$Y(p) = \frac{wa^*(p)}{\theta+(1-\theta)a^*(p)}\left[\frac{\theta}{\theta+(1-\theta)a^*(p)} + \frac{p(1-a)}{a}\right].$$

As $Y(p)$ is independent of σ, $\big|\partial^2 c_\theta(p)/\partial\theta\partial p\big| \propto \big|\xi_\theta'\big(T_f(p)\big)T_f'(p)\big|$. ■

To complete the proof, note that for each θ with $P_\theta > p$,

$$v_\theta := \frac{\sigma}{w\varphi\Big(\frac{T_f(p)-\lambda_1}{\sigma}\Big)}\frac{\partial c_\theta}{\partial p} = (\theta+(1-\theta)a^*(p))\left[1+\left[P_\theta\left(\frac{1}{2}-p\right)+\frac{p}{2}\right]\eta_a(p)\right]$$

$$\frac{\partial v_\theta}{\partial\theta} = (1-a^*(p))\left[1+\left[\frac{p-\frac{1}{2}}{a^*(p)-1}+\frac{p}{2}\right]\eta_a(p)\right].$$

Then,

$$\frac{\partial^2 c_\theta}{\partial p\partial\theta} > 0 \text{ if and only if } (1-a^*(p))\left[1+\left[\frac{p-\frac{1}{2}}{a^*(p)-1}+\frac{p}{2}\right]\eta_a(p)\right] > 0. \tag{1}$$

On the other hand, if $\theta = \theta(p)$, as $P_\theta = p$, $\partial c_\theta(p)/\partial p > 0$ if and only if $1 + p(1-p)\eta_a(p) > 0$. This value is positive, as shown in the following claim.

Claim 3 (Elasticity bound). *If $\partial a^*(p)/\partial p < 0$,*

$$-\frac{1}{(1-p)p} < -\frac{1}{2}\frac{2p-1}{(1-p)p} < \eta_a(p)$$

Proof of Claim 3. Recall that $a^*(p)$ is the solution to

$$\frac{K}{p-\kappa} = \frac{1}{1+a}\left[1-\tilde{G}_1(\omega(p,a))\right] + \frac{a}{1+a}\left[1-\tilde{G}_0(\omega(p,a))\right]$$

with respect to a. If $\partial a^*(p)/\partial p < 0$, $\omega(p,a) = 2\ln(a^*(p)) + \ln(p/(1-p))$ should increase in p. Otherwise, the right-hand side increases while the left-hand side decreases, a contradiction. Then, $\partial\omega(p,a)/\partial p > 0$ if and only if

$$-\frac{1}{2}\frac{2p-1}{(1-p)p} < \eta_a(p)$$

The last inequality follows from $p \leq 1$. ■

Recall that $\xi_\theta\big(T_f(p)\big) \to 0$ as $\sigma \to 0$ by Claim 1. Note that by a calculation,

$$\frac{\partial\Lambda(\lambda,p)}{\partial p} = -f\big(c_{\theta(p)}(p)\big)\frac{\partial c_{\theta(p)}(p)}{\partial p}(G_0(\theta(p)) - G_1(\theta(p)))$$

$$-\int_{\theta(p)}\left[f(c_\theta(p))\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} + f'(c_\theta(p))\frac{\partial c_\theta(p)}{\partial\theta}\frac{\partial c_\theta(p)}{\partial p}\right](G_0(\theta) - G_1(\theta))\,\mathrm{d}\theta.$$

By using Claim 2, for each $\theta > \theta(p)$, we have

$$\frac{\frac{\partial c_\theta(p)}{\partial p}\frac{\partial c_\theta(p)}{\partial\theta}}{\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p}} \to 0 \text{ as } \sigma \to 0.$$

Therefore, $\partial\Lambda(\lambda,p)/\partial p < 0$ if

$$\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} > 0 \text{ for } \theta > \theta(p) \text{ and } \frac{\partial c_{\theta(p)}(p)}{\partial p} > 0.$$

The latter inequality always holds by Claim 3. Thus, by equation (1), we have that $\partial\Lambda(\lambda,p)/\partial p < 0$ if

$$(1-a^*(p))\left[1+\left[\frac{p-\frac{1}{2}}{a^*(p)-1}+\frac{p}{2}\right]\eta_a(p)\right]>0.$$

■

Proof of Lemma 2. Note that the profit is written as

$$\Pi_\lambda(p)=\frac{\lambda}{2\sigma^3}\int_0^{a^*(p)}\Pi^*(p,a)\left[\varphi\left(\tfrac{\sigma}{\lambda}\ln(a)+\tfrac{\lambda}{2\sigma}\right)+\varphi\left(\tfrac{\sigma}{\lambda}\ln(a)-\tfrac{\lambda}{2\sigma}\right)\right]a\,\mathrm{d}a.$$

and

$$\varphi\left(\frac{\sigma}{\lambda}\ln(a)+\frac{\lambda}{2\sigma}\right)=\exp\left(-\tfrac{1}{2}\left(\tfrac{\sigma}{\lambda}\right)^2(\ln a)^2-\tfrac{1}{8}\left(\tfrac{\lambda}{\sigma}\right)^2\right)a^{-\frac{1}{2}}$$

Then, by letting $\rho=\lambda/\sigma$ and $W_\rho(a)=\exp\left(-\rho^{-2}(\ln a)^2/2-\rho^2/8\right)\left[a^{1/2}+a^{3/2}\right]$, the expected profit is written as $\Pi_\lambda(p)=\lambda\left(2\sigma^3\right)^{-1}\int_0^{a^*(p)}\Pi^*(p,a)W_\rho(a)\,\mathrm{d}a$.

The following claim shows that at the limit $\sigma\to 0$ or $\sigma\to\infty$ (equivalently, $\rho\to\infty$ or $\rho\to 0$), $W_\rho(\exp(\rho))$ dominates other $W_\rho(a)$.

Claim 4. *For any $a\in\mathbb{R}_+$, $W_\rho(a)/W_\rho(\exp(\rho))\to 0$ as $\rho\to 0$ or $\rho\to\infty$.*

Proof of Claim 4. By a calculation,

$$\frac{W_\rho(a)}{W_\rho(\exp(\rho))}=\exp\left(-\rho^{-2}\frac{(\ln(a))^2}{2}\right)\frac{\left[a^{\frac{1}{2}}+a^{\frac{3}{2}}\right]}{[\exp(\rho)]^{\frac{1}{2}}+[\exp(\rho)]^{\frac{3}{2}}}$$

This converges to 0 if either $\sigma\to 0$ or $\sigma\to\infty$. ■

Therefore, when $\sigma\to 0$ or $\sigma\to\infty$, $\Pi_\lambda(p)\to\lambda\left(2\sigma^3\right)^{-1}\Pi^*(p,\exp(\rho))$, which implies that $p^*_\sigma\to p_{a^\dagger_\sigma}$. ■

Proof of Proposition 1. Before the proof, we prepare the following claim:

Claim 5. *Suppose that the assumptions of Proposition 1 hold. Then, we have*

1. *When* $\sigma \to 0$, $p_\sigma^\dagger \to 1$.
2. $\lim_{\sigma\to 0} \omega(p_\sigma^\dagger, a^*(p_\sigma^\dagger)) < \infty$ *and* $a^*(p_\sigma^\dagger) \to 0$.
3. $\lim_{\sigma\to 0} \eta_a(p_\sigma^\dagger)(1-p_\sigma^\dagger)p_\sigma^\dagger = -1/2$.

Proof of Claim 5. Note that

$$\frac{\partial \Pi^*(p,a)}{\partial p} = \sum \pi_v \left(1 - \tilde{G}_v(\omega(p,a)) - (p-\kappa)\frac{\tilde{g}_v(\omega(p,a))}{p(1-p)}\right)$$

$$\frac{\partial \Pi^*(p,a)}{\partial a} = \frac{1}{a}\left[\pi_0\pi_1\big(\tilde{G}_1(\omega(p,a)) - \tilde{G}_0(\omega(p,a))\big) - 2(p-\kappa)\sum_v \pi_v \tilde{g}_v(\omega(p,a))\right],$$

where, $\pi_1 = 1/(1+a)$, and $\pi_0 = a/(1+a)$.

Note that the optimal price should be larger than $K > 0$. Suppose by contradiction that $\lim_{\sigma\to 0} p_\sigma^\dagger < 1$. Then, as $\omega(p,a) = 2\ln(a) + \ln(p/(1-p))$ and $a_\sigma^\dagger \to \infty$, $\omega(p,a) \to \infty$. Then, by the assumption that $\lim_{x\to\infty}\big(1-\tilde{G}_v(x)\big)/(\tilde{g}_v(x)) = \infty$, $\partial\Pi^*(p,a)/\partial p > 0$ for any p. This implies that $p_\sigma^\dagger \to 1$.

Next, consider $\omega(p_\sigma^\dagger, a^*(p_\sigma^\dagger))$. If $\omega(p_\sigma^\dagger, a^*(p_\sigma^\dagger)) \to \infty$, $\Pi(p_\sigma^\dagger, a^*(p)) \to 1-K > 0$, which is a contradiction. Furthermore, as $\lim_{\sigma\to 0}\omega(p_\sigma^\dagger, a^*(p_\sigma^\dagger)) < \infty$ and $p_\sigma^\dagger \to 1$, $a^*(p_\sigma^\dagger) \to 0$.

Finally, consider the elasticity of $a^*(p)$, which is calculated by

$$\eta_a(p) = \frac{\frac{\partial a^*(p)}{\partial p}}{a^*(p)} = -\frac{1}{a}\frac{\frac{\partial \Pi^*(p,a)}{\partial p}}{\frac{\partial \Pi^*(p,a)}{\partial a}}$$

$$= -\frac{\sum \pi_v\left(1 - \tilde{G}_v(\omega(p,a)) - (p-\kappa)\frac{\tilde{g}_v(\omega(p,a))}{p(1-p)}\right)}{\pi_0\pi_1\big(\tilde{G}_1(\omega(p,a)) - \tilde{G}_0(\omega(p,a))\big) - 2\sum_v \pi_v(p-\kappa)\tilde{g}_v(\omega(p,a))}.$$

Note that $\pi_0 \to 0$ as $a^*(p_\sigma^\dagger) \to 0$. Then, as $\lim_{\sigma\to 0}\omega(p_\sigma^\dagger, a^*(p_\sigma^\dagger)) < \infty$,

$$\eta_a(p)p(1-p) = -\frac{\sum \pi_v\big[p(1-p)\big(1-\tilde{G}_v(\omega(p,a))\big) - (p-\kappa)\tilde{g}_v(\omega(p,a))\big]}{\pi_0\pi_1\big(\tilde{G}_1(\omega(p,a)) - (p-\kappa)\tilde{G}_0(\omega(p,a))\big) - 2(p-\kappa)\sum_v \pi_v\tilde{g}_v(\omega(p,a))}$$

$$\to -\frac{-(p-\kappa)\sum \pi_v\tilde{g}_v(\omega(p,a))}{-2(p-\kappa)\sum_v \pi_v\tilde{g}_v(\omega(p,a))} \quad \text{(since } p \to 1 \text{ and } \pi_0 \to 0\text{)}.$$

Therefore, $\lim_{\sigma\to 0} p_\sigma^\dagger(1-p_\sigma^\dagger)\eta_a(p_\sigma^\dagger) = -1/2$. ■

Now we complete the proof. We use the sufficient condition shown in Theorem 1. When $a^*(p) < 1$ and $\partial a^*(p)/\partial p < 0$,

$$(1-a^*(p))\left[1+\left[\frac{p-\frac{1}{2}}{a^*(p)-1}+\frac{p}{2}\right]\eta_a(p)\right] = 1-a^*(p)+\left[\frac{1}{2}(1-p)-\frac{1}{2}pa^*(p)\right]\eta_a(p)$$

$$> 1-a^*(p)+\frac{1}{2}(1-p)\eta_a(p).$$

As $\lim_{\sigma\to 0}\eta_a(p_\sigma^\dagger)(1-p_\sigma^\dagger) = -1/2$, and $a^*(p_\sigma^\dagger) \to 0$ as $\sigma \to 0$, at $p = p_\sigma^\dagger$

$$\lim_{\sigma\to 0}\left[1-a^*(p)+\frac{1}{2}(1-p)\eta_a(p)\right] = \frac{3}{4} > 0.$$

Then, by Theorem 1, $\lim_{\sigma\to 0}\partial\Lambda(\lambda,p)/\partial p|_{p=p_\sigma^\dagger} < 0$. ■

A.2. Proofs in Section 4

Proof of Lemma 3. To simplify the notation, in this proof, we use T as the firm's cutoff. Note that the cutoff depends on the consumers' behavior. Therefore, we consider the cases (i) $t \geq T_H(p)$, (ii) $t \in (T_L(p), T_H(p))$, and (iii) $t \leq T_L(p)$.

(i) Suppose that $t \geq T_H(p)$. Note that if $t > T_H(p)$, the firm discontinues the sale as no consumer purchases the product, which leads to a deficit. Consider the case $t = T_H(p)$. Then, as only type-H consumers purchase the product, the firm's expected profit when the public signal t is observed is $\Pi_H(t)$, which is defined by

$$\Pi_H(t) := w\cdot([\Pr(v=1\mid t)\cdot\mu + \Pr(v=0\mid t)\cdot(1-\mu)]\cdot(p-\kappa)-K).$$

The condition when $\Pi_H(t) > 0$ is summarized as follows.

Claim 6. *(1) If $p-\kappa > K/(1-\mu)$, $\Pi_H(t) > 0$. (2) If $K/\mu > p-\kappa$, $\Pi_H(t) < 0$. (3) If $K/(1-\mu) > p-\kappa > K/\mu$, $\Pi_H(t) > 0$ if and only if $t < \tilde{T}$, where*

$$\tilde{T} := \frac{\lambda_0+\lambda_1}{2} + \frac{\sigma^2}{2}\frac{\ln\left(\frac{\mu-\frac{K}{p-\kappa}}{\frac{K}{p-\kappa}-(1-\mu)}\right)}{\lambda_0-\lambda_1}.$$

Using this claim, $T = T_H(p)$ if either (a) $p-\kappa > K/(1-\mu)$, or (b) $K/(1-\mu) > p-\kappa > K/\mu$ and $\tilde{T} \geq T_H(p)$. Otherwise, at $t = T_H(p)$, the profit is negative, and therefore, the cutoff is below $T_H(p)$.

(ii) Suppose that $T_L(p) < t < T_H(p)$. Then, at $t = T$, only the type-L consumer purchases the product. In this case, the firm's profit is $\Pi_H(t)$. Therefore, if $K/(1-\mu) > p-\kappa > K/\mu$ and $\tilde{T} \in (T_L(p), T_H(p))$, $T = \tilde{T}$. Otherwise, there is no t such that $\Pi_H(t) = 0$.

(iii) and (iv) Suppose that $t \leq T_L(p)$. Then, the firm's profit is $P_L := w[(p-\kappa)-K]$. Therefore, $P_L \geq 0$ if and only if $p-\kappa \geq K$.

The remaining cases to consider the cutoff point are the following.

(a) $p-\kappa \geq K/\mu$ and $T_L(p) \geq \tilde{T}$. Then, the profit is positive if and only if $t < T_L(p)$, the cutoff is $T_L(p)$.

(b) $K/\mu > p-\kappa \geq K$. Then, the profit is positive if and only if $t < T_L(p)$, and the cutoff is also $T_L(p)$.

(c) $K > p-\kappa$. Then, the profit is negative for any t. Therefore, the firm discontinues the sale; in other words, the cutoff is $-\infty$. ■

A.2.1. Proof of Proposition 2

Before the proof of Proposition 2, we prepare the following claim.

Claim 7. *Define $\underline{p}$ to be the largest solution to the following equation with respect to p:*

$$\frac{1-\mu}{\mu}\frac{1-p}{p} = \frac{\mu-\frac{K}{p-\kappa}}{\frac{K}{p-\kappa}-(1-\mu)}.$$

Then, $T_f(p) = T_H(p)$ if and only if $p > \overline{p}$ and $T_f(p) > T_L(p)$ if and only if $p > \underline{p}$. Further, $K/(1-\mu) + \kappa > \overline{p} > \underline{p} > K/\mu + \kappa$.

Proof of Claim 7. (a) If $p \geq K/(1-\mu) + \kappa$, $T_f(p) = T_H(p)$ by Lemma 3. If $p < K/(1-\mu) + \kappa$, $\tilde{T} < T_H(p)$ if and only if

$$\frac{\mu - \frac{K}{p-\kappa}}{\frac{K}{p-\kappa} - (1-\mu)} < \frac{\mu}{1-\mu}\frac{1-p}{p},$$

which is equivalently $p > \overline{p}$. Further, a simple calculation shows $K/(1-\mu) + \kappa > \overline{p}$.

(b) If $p < K/\mu + \kappa$, by Lemma 3, $T_f(p) = -\infty$. Consider the case where $p \geq K/\mu + \kappa$. Then, $T_f(p) = \max\{T_L(p), \tilde{P}\}$. Note that $T_L(p) < \tilde{T}$ if and only if

$$\frac{\mu - \frac{K}{p-\kappa}}{\frac{K}{p-\kappa} - (1-\mu)} > \frac{1-\mu}{\mu}\frac{1-p}{p},$$

equivalently, $g(p) > 0$, where g is a convex quadratic function (illustrated in Figure 3). Note that $g(p) < 0$ at $p = K/\mu + \kappa$. Then, $T_L(p) < T_f(p)$ if and only if $p > \underline{p}$. ■

Proof of Proposition 2. (1) Suppose that $p \geq \overline{p}$. If $p \geq \overline{p}$, by Claim 7, $T_f(p) = T_H(p)$. This implies that $c_\theta = -w \cdot \left[P_\theta\left(T_f(p)\right) - p\right]_+ \cdot \xi_\theta\left(T_f(p)\right) = 0$ for each $\theta \in \{\theta_H, \theta_L\}$. Therefore, $\lambda = \lambda_0 - \lambda_1 = (2\mu - 1) \cdot (F(c_L) - F(c_H)) = 0$, which implies that there is no informative equilibrium.

(2) Consider the case $p \in \left(\underline{p}, \overline{p}\right)$. Then, if $\lambda_0 > \lambda_1$, $T_L(p) < T_f(p) < T_H(p)$ for any c_H, c_L.

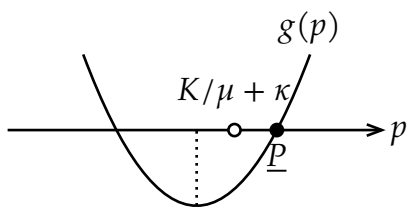

Figure 3: Illustration of g

We can easily verify that $\Lambda(\lambda, p) := \lambda_0 - \lambda_1 = (2\mu - 1) \cdot (F(c_L) - F(c_H))$ is continuous.

Below, we show that $\Lambda(\lambda, p) > \lambda$ for some $\lambda > 0$ when w is large enough. By the definition of $\overline{p}$ and $\underline{p}$, $P_H\big(T_f(p)\big) > p$, and $p > P_L\big(T_f(p)\big)$. This implies that

$$c_H = -w\big[P_H\big(T_f(p)\big) - p\big] \cdot \xi_H\big(T_f(p)\big), \text{ and } c_L = 0.$$

Then,

$$c_L - c_H = w \cdot \big[P_H\big(T_f(p)\big) - p\big] \cdot \xi_H\big(T_f(p)\big) > 0.$$

Fix $\overline{\lambda}$ so that $0 < \overline{\lambda} < (2\mu - 1)F(0)$. Then,

$$(2\mu - 1)(F(c_L) - F(c_H)) > \overline{\lambda} \Leftrightarrow F^{-1}\left(F(0) - \frac{\overline{\lambda}}{2\mu - 1}\right) > c_H.$$

Now take w so that

$$\frac{-F^{-1}\left(F(0) - \frac{\overline{\lambda}}{2\mu-1}\right)}{\big[P_H\big(T_f(p)\big) - p\big] \cdot \xi_H\big(T_f(p)\big)} < w.$$

Then, $\Lambda\big(\overline{\lambda}, p\big) > \overline{\lambda}$. The intermediate value theorem implies the existence of a fixed point of Λ, which is in $\big(\overline{\lambda}, 2\mu - 1\big)$. This implies that the fixed point is a positive value and shows the existence of an informative equilibrium. Moreover the equilibrim $\lambda^* \geq (2\mu - 1)F(0) - \delta$ for arbitorary $\delta \geq 0$ by taking $\overline{\lambda}$ close to $(2\mu - 1)F(0)$.

(3) Consider the case $p < \underline{p}$. Then, by Lemma 3 and Claim 7, if $p - \kappa > K$, $T_f(p) = T_L(p)$, and otherwise, $T_f(p) = -\infty$.

Suppose that $p - \kappa > K$. Then, by the definition of $T_L(p)$,

$$c_H = -w[P_H(T_L(p)) - p] \cdot \xi_H\big(T_f(p)\big), \text{ and } c_L = 0.$$

A similar way to case (2) shows the existence of an informative equilibrium.

If $p - \kappa \leq K$, as $T_f(p) = -\infty$. Then, the equilibrium becomes uninformative. ■

A.2.2. Proof of Proposition 3

Before we prove Proposition 3, we provide the following claim, which immediately follows from the definition of $\overline{p}$.

Claim 8. *$\mu > \overline{p}$ if and only if $(\mu - \kappa)/2 > K$.*

Now, we proceed to the proof of Proposition 3. Suppose that $(\mu - \kappa)/2 > \max\{K, 1 - \mu - \kappa\}$. This implies that $\mu > (2 - \kappa)/3$. Note that if $p = \mu$, as $(\mu - \kappa)/2 > K$, $p > \overline{p}$ by Claim 8. By Proposition 2, the equilibrium is uninformative and guarantees the profit $(\mu - \kappa)/2 - K$.

Below, we examine the existence of an informative equilibrium that yields a profit larger than $(\mu - \kappa)/2 - K$.

The expected profit is

$$\Pi(p) = \sum_{v' \in \{0,1\}} \frac{1}{2}\Big[\Phi\Big(\tfrac{T_f(p) - \lambda^*_{v'}(p)}{\sigma}\Big) \cdot \big[\big(v'\mu + (1 - v')(1 - \mu)\big)(p - \kappa) - K\big]$$

$$+\Phi\Big(\tfrac{T_L(p) - \lambda^*_{v'}(p)}{\sigma}\Big) \cdot \big((1 - v')\mu + v'(1 - \mu)\big)(p - \kappa)\Big].$$

Let $r = \lambda/(2\sigma)$. From Lemma 3, at the informative equilibrium, $T_L(p) \leq T_f(p) < T_H(p)$, and then, there are x_L, x_f such that for each $b \in \{L, H, f\}$, $(T_b(p) - \lambda_1)/\sigma = r + x_b$ and $(T_b(p) - \lambda_0)/\sigma = -r + x_b$, where $x_L < x_f < x_H$. Furthermore, as $p > \kappa + K > 1/2$ by Proposition 2,

$$x_L = \ln\Big(\frac{1 - \mu}{\mu}\frac{1 - p}{p}\Big)\frac{1}{2r} < x_f < x_H = \ln\Big(\frac{\mu}{1 - \mu}\frac{1 - p}{p}\Big)\frac{1}{2r} < -x_L.$$

Then, $x_L < -x_f$ implies that $\Phi(x_L + r) \leq \Phi\big(-x_f + r\big)$ and $\Phi(x_L - r) \leq \Phi\big(-x_f - r\big)$. By using this inequality and the symmetry of Φ,

$$\Pi(p) < \frac{\Phi\big(x_f + r\big) + \Phi\big(-x_f - r\big)}{2}\mu(p-\kappa) + \frac{\Phi\big(x_f - r\big) + \Phi\big(-x_f + r\big)}{2}(1-\mu)(p-\kappa)$$

$$-\frac{K}{2}\Big[\Phi\big(x_f + r\big) + \Phi\big(x_f - r\big)\Big]$$

$$= \frac{p-\kappa}{2} - \frac{K}{2}\Big[\Phi\big(x_f + r\big) + \Phi\big(x_f - r\big)\Big].$$

Furthermore,

$$x_f = \ln\left(\max\left\{\frac{1-\mu}{\mu}\frac{1-p}{p}, \frac{\mu - \frac{K}{p-\kappa}}{\frac{K}{p-\kappa} - (1-\mu)}\right\}\right)$$

If $p > \kappa + 2K$, $x_f > 0$. Then, $\Phi\big(x_f + r\big) + \Phi\big(x_f - r\big)$ is decreasing in r and thus, no less than 1. Therefore, $\Pi(p) < (p - \kappa - K)/2$. If $p < \kappa + 2K$, the maximum profit is K.

As $p < \overline{p}$, the maximum profit is

$$\frac{\overline{p} - \kappa - K}{2} = \frac{K}{2}\frac{\mu + \kappa(2\mu - 1)}{\mu(1-\mu) + (2\mu - 1)K} - \frac{K}{2}.$$

Therefore, uninformative profit exceeds it if $\mu > 4K + \kappa$ and

$$\mu - \kappa > K\frac{\mu + \kappa(2\mu - 1)}{\mu(1-\mu) + (2\mu - 1)K} + K.$$

A.2.3. Proof of Proposition 4

The proof consists of the following two claims, which hold under the assumption of Proposition 4.

Claim 9. *If the price induces an uninformative equilibrium, the expected profit is at most* 0.

Claim 10. *By setting $p = \mu$, the equilibrium is informative. The expected profit is no less than $(\mu - \kappa - K)/2$ in this case.*

Proof of Claim 9. By Proposition 2, the situation is either $p > \overline{p}$ or $p < \kappa + K$. From Claim 8 and assumption $(\mu - \kappa)/2 - K < 0$, $\overline{p} > \mu$. If $p > \overline{p} > \mu$, as the ex post belief of type H, $P_H = \mu < p$ at the uninformative equilibrium, the profit is 0 as the type-H

never purchases. At the uninformative equilibrium with $p < \kappa + K$, the profit is also 0 as the firm gives up selling. ■

Proof of Claim 10. From Claim 8, $K > (\mu - \kappa)/2$ implies that $\overline{p} > \mu > 1 - \mu$, and as $\mu > K + \kappa$, Proposition 2 implies that an equilibrium exists and it is necessarily informative at $p = \mu$.

Recall that the expected profit is

$$\Pi(p) = \sum_{v' \in \{0,1\}} \frac{1}{2}\Big[\Phi\Big(\tfrac{T_f(p) - \lambda^*_{v'}(p)}{\sigma}\Big) \cdot \big[\big(v'\mu + (1 - v')(1 - \mu)\big)(p - \kappa) - K\big]$$

$$+\Phi\Big(\tfrac{T_L(p) - \lambda^*_{v'}(p)}{\sigma}\Big) \cdot \big((1 - v')\mu + v'(1 - \mu)\big)(p - \kappa)\Big].$$

By Proposition 2, we can take λ as a positive value by taking w large. Then, by $\sigma \to 0$,

$$\Pi(\mu) \to \frac{\mu - \kappa - K}{2}.$$

■

Proof of Proposition 4. By Claims 9 and 10, and assumption $\mu > \kappa + K$, a price that leads to an informative equilibrium yields a positive profit, which is larger than the profit of any uninformative equilibrium. Then, the optimal price leads to an informative equilibrium. ■

A.3. Proofs in Section 5

Proof of Lemma 4. First note that for sufficiently small t, $\Pi^*_L(t) > \Pi^*_H(t)$. This is because for sufficiently small $t \approx -\infty$, $P_\theta(t) \approx 1$ and therefore $\Pi^*_L(t) \approx 1 > \mu \approx \Pi^*_H(t)$.

Recall that $P_\theta(t)$ is written as $P_\theta(t) = 1/(1 + a(1 - \theta)/\theta)$, where $a = \exp\big((\lambda_0 - \lambda_1)\sigma^{-2}(t - (\lambda_0 + \lambda_1)/2)\big)$ and also $\Pr(v = 1 \mid t) = 1/(1 + a)$. To simplify the notation, let denote $P(t) = \Pr(v = 1 \mid t)$. By differentiation, if $\lambda_0 > \lambda_1$, we have

$$P'_\theta(t) = -\frac{\lambda_0 - \lambda_1}{\sigma^2} \cdot \frac{1-\theta}{\theta} \cdot a \cdot (P_\theta(t))^2 < 0.$$

Note that we can write $P_H(t) = 1/(1 + a(1-\mu)/\mu)$ and $P_L(t) = 1/(1 + a\mu/(1-\mu))$. This implies that $P_H(t)/P_L(t)$ and $P(t)/P_L(t)$ are increasing in t.

Now we calculate the ratio of the profits:

$$\begin{aligned} \frac{\Pi_H^*(t)}{\Pi_L^*(t)} &= (P(t)(2\mu - 1) + (1-\mu))\frac{P_H(t)}{P_L(t)} > 1 \iff Z(a) > 0, \\ Z(a) &:= (1-\mu)\mu^2 a^2 - (2\mu - 1)(1 - (1-\mu)\mu)a - \mu(1-\mu)^2 > 0. \end{aligned} \tag{2}$$

Note that $Z(a)$ is a convex quadratic function of a, and $Z(0) < 0$. Also note that as $P_L(t) > K, a < (1/K - 1) \cdot (1-\mu)/\mu$. A calculation shows that $Z((1/K-1) \cdot (1-\mu)/\mu)$ is decreasing in μ, and let $\mu^\dagger$ be the solution that $Z((1/K-1) \cdot (1-\mu)/\mu) = 0$. This implies that if $\mu < \mu^\dagger$, $\Pi_H^*(t) < \Pi_L^*(t)$ for each t. On the contrary, if $\mu > \mu^\dagger$, there is $\hat{t} < T_L(K)$ such that $\Pi_H^*(t) < \Pi_L^*(t)$ if and only if $t < \hat{t}$.

(c) As $T_L(K)$ is the solution of $P_L(t) = K$, That is,

$$T_L(K) = \frac{\lambda_0 + \lambda_1}{2} + \frac{\sigma^2}{\lambda_0 - \lambda_1} \cdot \ln\Big(\big(\tfrac{1}{K} - 1\big) \cdot \tfrac{1-\mu}{\mu}\Big).$$

Let μ^* be the solution to $(1/K - 1) \cdot (1-\mu)/\mu = 1$. This exists if $K < 1/2$. Then, if $\mu < \mu^*$, $T_L(K) \to \infty$. In contrast, if $\mu > \mu^*$, $T_L(K) \to -\infty$. We can also verify that $Z((1/K - 1) \cdot (1-\mu^*)/\mu^*) < 0$, that is, $Z(1) < 0$. This implies that $\mu^* < \mu^\dagger$. ■

A.3.1. Proof of Theorem 3

Proof of case (a): We first consider the high μ case: there is $\hat{t} < T_L(K)$ such that $\Pi_L^*(t) > \Pi_H^*(t)$ if and only if $t < \hat{t}$.

Let T^* be the solution to $\Pi_H^*(t) = 0$ with respect to t. This implies that at the optimal strategy, the firm sets price $P_L(t)$ when $t \in (-\infty, \hat{t})$, sets price $P_H(t)$ when $t \in (\hat{t}, T^*)$, and when $t > T^*$, discontinues the product sale.

Under this firm's behavior, the expected utility of the type-H is

$$w\left[\int_{\hat{t}-\mathrm{d}\tau}^{T^*-\mathrm{d}\tau}[P_H(t)-P_H(t+\mathrm{d}\tau)]\xi_H(t)\,\mathrm{d}t+\int^{\hat{t}-\mathrm{d}\tau}[P_H(t)-P_L(t+\mathrm{d}\tau)]\cdot\xi_H(t)\,\mathrm{d}t\right].$$

Here, $P_H(t)$ is the expected value of the product for type-H while the price is set to $p = P_L(t)$. A marginal increase in turnout $\mathrm{d}\tau$ affects the price. As the firm sets price $p = P_H(t)$ if $t > \hat{t}$, type-H consumers' informational rent $P_H(t) - P_L(t)$ vanishes. The differentiation concerning $\mathrm{d}\tau$ is the marginal gain by participation. Therefore, type-H consumer i participates in the protest if and only if $c_H \geq c_i$, where

$$c_H := w\cdot\left[\underbrace{-\big[P_H(\hat{t})-P_L(\hat{t})\big]\xi_H(\hat{t})}_{\text{information rent}}+\underbrace{\int_{\hat{t}}^{T^*}\big[-P_H'(t)\big]\xi_H(t)\,\mathrm{d}t+\int^{\hat{t}}\big[-P_L'(t)\big]\xi_H(t)\,\mathrm{d}t}_{\text{benefit from price reduction by participation}}\right].$$

Now we compare with the type-L consumers' utility. The utility of the type-L is $w\cdot\int^{\hat{t}-\mathrm{d}\tau}[P_L(t)-P_L(t+\mathrm{d}\tau)]\cdot\xi_L(t)\,\mathrm{d}t$ as the firm extracts type-L consumers' full surplus. Then, the type-L consumers participate in the protest if and only if

$$c_L := w\cdot\int^{\hat{t}}\big[-P_L'(t)\big]\xi_L(t)\,\mathrm{d}t \geq c_i.$$

Then, $c_L > c_H$ if and only if

$$D := \int^{\hat{t}}\big[-P_L'(t)\big](\xi_L(t)-\xi_H(t))\,\mathrm{d}t-\int_{\hat{t}}^{T^*}\big[-P_H'(t)\big]\xi_H(t)\,\mathrm{d}t+\big[P_H(\hat{t})-P_L(\hat{t})\big]\xi_H(\hat{t}),$$

which is shown to be negative.

Claim 11. *Suppose that $\mu > \max\{\mu^\dagger, 2K\}$. Then, $D < 0$ for sufficiently large σ, and $D > 0$ for sufficiently small σ.*

Proof of Claim 11. Recall that $a = \exp\big((\lambda_0-\lambda_1)\sigma^{-2}\cdot[t-(\lambda_1+\lambda_0)/2]\big)$ and P_θ is written as $P_\theta = 1/(1+a(1-\theta)/\theta)$. Then,

$$P_H(t) - P_L(t) = \frac{\frac{\mu}{1-\mu} - \frac{1-\mu}{\mu}}{\left(1 + \frac{\mu}{1-\mu}a\right) \cdot \left(1 + \frac{1-\mu}{\mu}a\right)} a,$$

$$-P'_\theta(t) = \frac{\lambda_0 - \lambda_1}{\sigma^2} \frac{1-\theta}{\theta} \cdot a \cdot (P_\theta(t))^2 > 0.$$

$$\xi_L(t) - \xi_H(t) = \frac{2\mu - 1}{\sigma} \cdot \varphi\left(\frac{t-\lambda_1}{\sigma}\right) \cdot [a - 1].$$

As shown in the proof of Lemma 4, $\Pi^*_H(t) = \Pi^*_L(t)$ if and only if $Z(a) = 0$. (Z is defined in (2)). Therefore, when we write $\hat{a} = \exp\big((\lambda_0 - \lambda_1)\sigma^{-2}(\hat{t} - (\lambda_1 + \lambda_0)/2)\big)$, $Z(\hat{a}) = 0$. Also, this implies that $P_H(\hat{t}) - P_L(\hat{t})$ is independent of σ.

We first consider a large σ case. Then, even $\sigma \to \infty$, $\hat{a}$ is a finite value, which implies that $\hat{t}/\sigma^2$ converges to a finite real number. Further,

$$\frac{\xi_H(t)}{\xi_H(\hat{t})} = \frac{\mu \cdot \exp\left((\hat{t} - t) \cdot \frac{\hat{t}+t-2\lambda_1}{2\sigma^2}\right) + (1-\mu) \cdot \exp\left((\hat{t} - t + \lambda_0 - \lambda_1) \cdot \frac{\hat{t}+t-(\lambda_0+\lambda_1)}{2\sigma^2}\right)}{\mu + (1-\mu) \cdot \hat{a}}.$$

As $\hat{t}/\sigma^2$ converges to a real number, for each t with $|t| < |\hat{t}| + z$ for some constant z, $\xi_H(t)/\xi_H(\hat{t}) \to \infty$. In contrast, if $t < \hat{t}$, $\xi_H(t)/\xi_H(\hat{t}) \to 0$ as $\hat{t} \to -\infty$. By Lemma 4, as $\mu > \mu^\dagger > \mu^*$, $T_L(K) \to -\infty$ and $\hat{t} < T_L(K)$. Then, $\hat{t} \to -\infty$ as $\sigma \to \infty$.

Dividing D by $\xi_H(\hat{t})$ yields that

$$\int^{\hat{t}} [-P'_L(t)] \frac{\xi_L(t) - \xi_H(t)}{\xi_H(\hat{t})} \, \mathrm{d}t - \int_{\hat{t}}^{T^*} [-P'_H(t)] \frac{\xi_H(t)}{\xi_H(\hat{t})} \, \mathrm{d}t + [P_H(\hat{t}) - P_L(\hat{t})]. \tag{3}$$

Note also that $\xi_L(t) < \xi_H(t)$ for sufficiently small t when $\lambda_0 > \lambda_1$. Then, the first term is negative as $P'_L(t) < 0$. Also note that by $\mu/2 > K$, $T^* \to \infty$ as $\sigma \to \infty$. Therefore, as $\sigma \to \infty$, the value of (3) is lower than

$$\int_{-\infty}^{\infty} P'_H(t) \frac{\xi_H(t)}{\xi_H(\hat{t})} \, \mathrm{d}t + [P_H(\hat{t}) - P_L(\hat{t})].$$

As $[P_H(\hat{t}) - P_L(\hat{t})]$ is indepdendent of σ, and $\xi_H(t)/\xi_H(\hat{t}) \to \infty$, the above equation diverges to $-\infty$. This shows that $D < 0$.

Next, consider a low σ case. Since $\Pi_H^*(t)$ can be represented by σ-independent form using $a = \exp\big(\lambda\sigma^{-2}(t - (\lambda_0 + \lambda_1)/2)\big)$, $T^* = \sigma^2\lambda^{-1}\ln\big(a^H\big) + (\lambda_0 + \lambda_1)/2$ for some a^H, which is independent of σ. Also, since $\hat{t} = \sigma^2\lambda^{-1}\ln(\hat{a}) + (\lambda_0 + \lambda_1)/2$,

$$T^* - \hat{t} = \sigma^2\lambda^{-1}\ln\big(a^H/\hat{a}\big).$$

Further,

$$\frac{\xi_H(t)}{\xi_H(\hat{t})} = \frac{\mu \cdot \exp\Big(\big(\hat{t} - t\big) \cdot \frac{\hat{t}+t-2\lambda_1}{2\sigma^2}\Big) + (1-\mu) \cdot \exp\Big(\big(\hat{t} - t + \lambda_0 - \lambda_1\big) \cdot \frac{\hat{t}+t-(\lambda_0+\lambda_1)}{2\sigma^2}\Big)}{\mu + (1-\mu)\cdot\hat{a}}$$

$$\frac{\xi_L(t) - \xi_H(t)}{\xi_H(\hat{t})} = (2\mu - 1) \cdot \frac{\exp\Big(\big(\hat{t} - t\big) \cdot \frac{\hat{t}+t-2\lambda_1}{2\sigma^2}\Big) - \exp\Big(\big(\hat{t} - t + \lambda_0 - \lambda_1\big) \cdot \frac{\hat{t}+t-(\lambda_0+\lambda_1)}{2\sigma^2}\Big)}{\mu + (1-\mu)\cdot\hat{a}}$$

Note that both T^* and $\hat{t}$ converge to $(\lambda_0 + \lambda_1)/2$ as $\sigma \to 0$. Then, if $t \in \big(\hat{t}, T^*\big)$,

$$0 \le \big(\hat{t} - t + \lambda_0 - \lambda_1\big) \cdot \frac{\hat{t} + t - (\lambda_0 + \lambda_1)}{2\sigma^2} < \ln(\hat{a}) + \ln\big(a^H\big)$$

$$\big(\hat{t} - t\big) \cdot \frac{\hat{t} + t - 2\lambda_1}{2\sigma^2} \to 0.$$

Then, $\xi_H(t)/\xi_H\big(\hat{t}\big)$ converges to a finite value. Thus,

$$\int_{\hat{t}}^{T^*} \big[-P_H'(t)\big]\frac{\xi_H(t)}{\xi_H(\hat{t})}\,\mathrm{d}t < [T^* - t]\max_t\left[-P_H'(t)\frac{\xi_H(t)}{\xi_H(\hat{t})}\right] \to 0.$$

Similary, if $t < \hat{t}$,

$$\big(\hat{t} - t + \lambda_0 - \lambda_1\big) \cdot \big(\hat{t} + t - (\lambda_0 + \lambda_1)\big) \to \left(\frac{\lambda_0 + \lambda_1}{2} - t + \lambda\right)\left(t - \frac{\lambda_0 + \lambda_1}{2}\right) < 0$$

Therefore, $(\xi_L(t) - \xi_H(t))/\xi_H\big(\hat{t}\big)$ converges to a positive value as $\sigma \to 0$. Using (3), we can conclude that $D > 0$ for sufficiently small σ. ■

Proof of (b): Next, we consider the low μ case: for each $t < T_L(K)$, $\Pi_L^*(t) > \Pi_H^*(t)$. This implies that for any $t < T_L(K)$, the firm sets price $P_L(t)$ and discontinues the product sale if $t > T_L(K)$.

To simplify the notation, we denote $T_* = T_L(K)$. The utility of the type-H from the product sales is

$$w \int^{T_* - \mathrm{d}\tau} [P_H(t) - P_L(t + \mathrm{d}\tau)] \xi_H(t) \, \mathrm{d}t.$$

The differentiation concerning $\mathrm{d}\tau$ is the marginal gain by participation. Therefore, they participate in the protest if and only if

$$c_H := -w[P_H(T_*) - P_L(T_*)]\xi_H(T_*) + w \int^{T_*} \left[-P_L'(t)\right] \xi_H(t) \, \mathrm{d}t \geq c_i.$$

Similarly, the utility of the type-L is $w \int^{T_* - \mathrm{d}\tau} [P_L(t) - P_L(t + \mathrm{d}\tau)] \xi_L(t) \, \mathrm{d}t$. Then, they participate in the protest if and only if

$$c_L := w \int^{T_*} \left[-P_L'(t)\right] \xi_L(t) \, \mathrm{d}t \geq c_i.$$

Then, $c_L > c_H$ if and only if $\tilde{D} > 0$, which is defined by

$$\tilde{D} := \int^{T_*} \left[-P_L'(t)\right] (\xi_L(t) - \xi_H(t)) \, \mathrm{d}t + [P_H(T_*) - P_L(T_*)] \xi_H(T_*).$$

We complete the proof with the following claim.

Claim 12. *Suppose that $\mu \in \left(\mu^*, \mu^\dagger\right)$. Then, for sufficiently large σ, $\tilde{D} > 0$.*

Proof of Claim 12. Dividing $\tilde{D}$ by $\xi_H(T_*)$ yields that

$$\frac{\int^{T_*} \left[-P_L'(t)\right] (\xi_L(t) - \xi_H(t)) \, \mathrm{d}t}{\xi_H(T_*)} + [P_H(T_*) - P_L(T_*)] \tag{4}$$

As $\mu > \mu^*$, $\lim_{\sigma \to \infty} T_* = -\infty$ by Lemma 4. Therefore, the first term of (4) converges to

$$-P_L'(T_*) \frac{\xi_L(T_*) - \xi_H(T_*)}{\xi_H'(T_*)}.$$

Note that

$$\frac{\xi_L(T_*) - \xi_H(T_*)}{\xi_H'(T_*)} = \frac{(2\mu - 1)\left(1 - \exp\left(\frac{\lambda_0 - \lambda_1}{\sigma^2}\left(T_* - \frac{\lambda_1 + \lambda_0}{2}\right)\right)\right)}{\mu \frac{T_* - \lambda_1}{\sigma^2} + (1 - \mu) \frac{T_* - \lambda_0}{\sigma^2} \exp\left(\frac{\lambda_0 - \lambda_1}{\sigma^2}\left(T_* - \frac{\lambda_1 + \lambda_0}{2}\right)\right)}.$$

As shown in Lemma 4, T_*/σ^2 converges to a finite value. Then, the above value converges to a finite value.

Note also that

$$-P_L'(t) = \frac{\lambda_0 - \lambda_1}{\sigma^2}\frac{\mu}{1-\mu} \times \frac{1}{\frac{1}{a} + 2\frac{\mu}{1-\mu} + \left(\frac{\mu}{1-\mu}\right)^2 a} > 0.$$

As T_*/σ^2 converges to a finite value, a is also a finite value. Then, $-P_L'(T_*) \to 0$. Therefore, (4) reduced to $[P_H(T_*) - P_L(T_*)]$, which is calculated as

$$P_H(t) - P_L(t) = \frac{\frac{\mu}{1-\mu} - \frac{1-\mu}{\mu}}{\left(1 + \frac{\mu}{1-\mu}a\right)\left(1 + \frac{1-\mu}{\mu}a\right)}a,$$

As a is a finite value, $\lim_{\sigma^2 \to \infty} P_H(T_*) - P_L(T_*) > 0$. This concludes that $\tilde{D} > 0$. ■

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Online Appendix

B. Omitted proof for standard fact

Proof of Fact 1. Note that ex post belief $\{P_\theta(t)\}_{t\in\mathbb{R}}$ has mean θ, which is independent of λ.

Note also that $P_\theta(t) = \theta/(\theta + (1-\theta)a)$, where $a = \exp\big(\lambda\sigma^{-2}\cdot(t-(\lambda_0+\lambda_1)/2)\big)$. By letting $Z = \ln\big(\theta(1-\theta)^{-1}(1-\pi)\pi^{-1}\big)$,

$$\Pr(P_\theta(t) > \pi \mid \lambda) = \theta\Phi\Big(\tfrac{\lambda}{2\sigma} + \tfrac{\sigma}{\lambda}Z\Big) + (1-\theta)\Phi\Big(\tfrac{\lambda}{2\sigma} - \tfrac{\sigma}{\lambda}Z\Big)$$

and then,

$$\frac{\partial \Pr(P_\theta(t) > \pi \mid \lambda)}{\partial\lambda} > 0 \iff \frac{\pi - \frac{1}{2}}{\sigma^2}\lambda^2 + \ln\Big(\frac{\pi}{1-\pi}\Big) > \ln\Big(\frac{\theta}{1-\theta}\Big)$$

As the left-hand side of the above inequality is increasing in π, $\Pr(P_\theta(t) > \pi \mid \lambda)$ has the single-crossing property, which implies that $\Pr(P_\theta(t) > \pi \mid \lambda)$ with a higher λ is a mean-preserving spread of that with a lower λ. ■

C. High-noise case for continuous types

In this section, we analyze the case where σ is sufficiently high, and examine price-informativeness interaction in continuous signal model.

As we discussed in Section 3.2, the effect of p on c_θ are decomposed into the pivotal effect and the information-rent effect as follows:

$$c_\theta(p) = -w\underbrace{\big[P_\theta\big(T_f(p)\big) - p\big]_+}_{\text{Information rent}} \cdot \underbrace{\xi_\theta\big(T_f(p)\big)}_{\text{Pivotal prob.}}.$$

We show that the pivotal effect dominates the information-rent effect when σ is sufficiently high. First note that, when σ is high, $T_f(p)$ is sufficiently large. Because ξ_θ is the mean of $\varphi((t-\lambda_v)/\sigma)$ and $\varphi'(t') < 0$ for sufficiently large t', it follows that $\xi'_\theta(t') < 0$. Therefore, the probability of being pivotal increases. This widens the gap in c_θ across different θ. Since the informativeness of the protest depends on the difference in participation cutoffs across θ, λ increases as p increases. The effect on information rent

is dominated by the pivotal effect, because the effect on information rent is proportional to ξ_θ, which converges to zero at the rate $\sigma^{-1}\exp(-\sigma)$.

The followng theorem formally summarizes the above discussion.

Theorem 4. *$\partial\Lambda(\lambda,p)/\partial p > 0$ for sufficiently large σ if and only if $[\partial a^*(p)/\partial p]\cdot\ln(a^*(p)) < 0$.*

Now we consider the optimal price choice in a high-noise environment. Let $p_\sigma^\dagger$ be the optimal price when the noise level is σ when the effect on λ is ignored.

As Lemma 2 shows, in a high-noise environment (i.e., $\sigma \to \infty$), we have $\lim_{\sigma\to\infty} p_\sigma^\dagger = p_1$; that is, the price maximizes $\Pi^*(p,1)$. The case $a = 1$ corresponds to $t = 0$, which implies no information is obtained from the protest. If the firm prefers to sell a product (i.e., $\Pi^*(p_1,1) > 0$), then $a^*(p_1) > 1$. Therefore, by Theorem 4, $\partial\Lambda(\lambda,p_1)/\partial p_1 > 0$ for sufficiently large σ if and only if $\partial a^*(p_1)/\partial p_1 < 0$. $\partial a^*(p)/\partial p < 0$ is satisfied, for example, if $\partial^2\Pi^*(p,a)/\partial p\partial a < 0$. We provide a sufficient condition below. In this case, price and informativeness are complements: the optimal price should be higher when the firm prefers more informative protests.

Proposition 5. *Suppose that $g_1(\theta) = 2\theta$ and $g_0(\theta) = 2(1-\theta)$. Then, if $a^*(p_1) \approx 1$ and $\Pi^*(p_1,1) > 0$, $\lim_{\sigma\to\infty}\partial\Lambda\big(\lambda,p_\sigma^\dagger\big)/\partial p > 0$.*

C.1. Proofs

Proof of Theorem 4. We first consider the case with a high σ.

Note that we write that for any function of σ, $x(\sigma)$ and $y(\sigma)$, $x \propto y$ if $\lim_{\sigma\to\infty}|x(\sigma)/y(\sigma)| \in \mathbb{R}_{++}$. We prepare the following two claims.

Claim 13. *As $\sigma \to \infty$, $\left|\xi_\theta\big(T_f(p)\big)\right| \propto \sigma^{-1}e^{-\sigma}$ and $\big(\xi_\theta'\big(T_f(p)\big)\big)/\big(\xi_\theta\big(T_f(p)\big)\big) \to -\ln(a^*(p))$.*

Proof of Claim 13.

$$\xi_\theta'\big(T_f(p)\big) = -\frac{1}{\sigma}\varphi\Big(\tfrac{T_f(p)-\lambda_1}{\sigma}\Big)\Big[\theta\tfrac{T_f(p)-\lambda_1}{\sigma^2} + (1-\theta)\tfrac{T_f(p)-\lambda_0}{\sigma^2}a\Big],$$

$$T_f(p) - \lambda_1 = \frac{\lambda}{2} + \frac{\sigma^2}{\lambda}\ln(a^*(p)), \text{ and } T_f(p) - \lambda_0 = -\frac{\lambda}{2} + \frac{\sigma^2}{\lambda}\ln(a^*(p)).$$

$$\xi_\theta\big(T_f(p)\big) = \frac{1}{\sigma}\varphi\Big(\tfrac{T_f(p)-\lambda_1}{\sigma}\Big)[\theta + (1-\theta)a],$$

Then, $\lim_{\sigma\to\infty} \frac{\xi'_\theta\big(T_f(p)\big)}{\xi_\theta\big(T_f(p)\big)} = -\ln(a^*(p))$. ■

The next claim shows convergence rates of the derivatives of c_θ.

Claim 14. *For sufficiently large σ,*

$$\operatorname{sign}\left(\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p}\right) = \operatorname{sign}\left(\frac{\partial a^*(p)}{\partial p}\ln(a^*(p))\right).$$

Proof of Claim 14. As in the proof of Claim 2, the cross derivative is calculated as

$$\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} = -\xi'_\theta\big(T_f(p)\big)T'_f(p)Y(p) - \xi_\theta\big(T_f(p)\big)Y'(p),$$

$$Y(p) = \frac{wa^*(p)}{\theta + (1-\theta)a^*(p)}\left[\frac{\theta}{\theta + (1-\theta)a^*(p)} + \frac{p(1-a)}{a}\right].$$

As $\theta/(\theta + (1-\theta)a) > p$ and $(1-a)/a > -1$, $Y(p) > 0$. As $\lim_{\sigma\to\infty} \frac{\xi'_\theta\big(T_f(p)\big)}{\xi_\theta\big(T_{f(p)}\big)} = -\ln(a^*)$, for sufficiently large σ,

$$T'_f(p) = \frac{\sigma^2}{\lambda}\frac{\partial a^*(p)}{\partial p}, \text{and } \operatorname{sign}\left(\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p}\right) = \operatorname{sign}\left(\frac{\partial a^*(p)}{\partial p}\ln(a^*(p))\right).$$

■

Finally, we determine the sign of $\partial\Lambda(\lambda,p)/\partial p$. Note that

$$\begin{aligned}
\frac{\partial\Lambda(\lambda,p)}{\partial p} &= \int_{\theta(p)} f(c_\theta(p))\frac{\partial c_\theta(p)}{\partial p}(g_0(\theta) - g_1(\theta))\,\mathrm{d}\theta \\
&= -f\big(c_{\theta(p)}(p)\big)\frac{\partial c_{\theta(p)}(p)}{\partial p}(G_0(\theta(p)) - G_1(\theta(p))) \\
&\quad - \int_{\theta(p)}\left[f(c_\theta(p))\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} + f'(c_\theta(p))\frac{\partial c_\theta(p)}{\partial\theta}\frac{\partial c_\theta(p)}{\partial p}\right](G_0(\theta) - G_1(\theta))\,\mathrm{d}\theta
\end{aligned}$$

Claim 13 implies that $\xi_\theta\big(T_f(p)\big) \to 0$ and $\xi'_\theta\big(T_f(p)\big)/\xi_\theta\big(T_f(p)\big) \to -\ln(a^*(p))$ as $\sigma \to \infty$. Furthermore, we can see that $T'_f(p) \to \infty$ as $\sigma \to \infty$. By Claim 2, these facts imply that

$$\lim_{\sigma\to\infty} \frac{x}{\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p}} \to 0 \text{ for each } x \in \left\{ \frac{\partial c_\theta(p)}{\partial\theta}\frac{\partial c_\theta(p)}{\partial p}, \frac{\partial c_{\theta(p)}(p)}{\partial p} \right\}.$$

Therefore,

$$\frac{\partial\Lambda(\lambda,p)}{\partial p}\frac{\partial^2 c_\theta(p)}{\partial\theta\partial p} < 0$$

for sufficiently large σ. Then, by Claim 14, the desired result holds. ■

Proof of Proposition 5. By the first-order condition, at $p = p_a$,

$$\frac{\partial\Pi^*(p,a)}{\partial p} = \frac{1}{1+a}\big(1-\tilde{G}_1 + a(1-\tilde{G}_0)\big) - \frac{p-\kappa}{1+a}(\tilde{g}_1 + a\tilde{g}_0)\frac{1}{p(1-p)} = 0$$

Therefore, at $p = p_1$,

$$\frac{p-\kappa}{p(1-p)} = \frac{2-\tilde{G}_1-\tilde{G}_0}{\tilde{g}_1+\tilde{g}_0}$$

On the other hand,

$$\begin{aligned}(1+a)\frac{\partial^2\Pi^*(p,a)}{\partial p\partial a} &= \frac{1}{1+a}\big(\tilde{G}_1-\tilde{G}_0\big) - \frac{1}{2a}(\tilde{g}_1+a\tilde{g}_0) \\ &\quad + \frac{p-\kappa}{p(1-p)}\left[\frac{\tilde{g}_1-\tilde{g}_0}{1+a} - \frac{\tilde{g}_1'+a\tilde{g}_0'}{2a}\right].\end{aligned}$$

Therefore, $\partial^2\Pi^*(p,a)/\partial p\partial a < 0$ at $a = 1$ and $p = p_1$ if

$$A := \frac{1}{2}\big(\tilde{G}_1-\tilde{G}_0\big) - \frac{1}{2}(\tilde{g}_1+\tilde{g}_0) + \frac{2-\tilde{G}_1-\tilde{G}_0}{\tilde{g}_1+\tilde{g}_0}\left[\frac{\tilde{g}_1-\tilde{g}_0}{2} - \frac{\tilde{g}_1'+\tilde{g}_0'}{2}\right] < 0.$$

Now suppose that $g_1(\theta) = 2\theta$ and $g_0(\theta) = 2(1-\theta)$. Then, the above equation is calculated as follows.[17]

$$A = -\theta(1-\theta) - 1 + 2(1-\theta)(2\theta-1) < 0,$$

[17] Note that $\tilde{G}_v(x) = G_v(\psi(x))$. Then, as g_v is linear, $\tilde{g}_v = g_v\psi'$ and $\tilde{g}_v' = g_v\psi''$. Recall that $\psi = \exp(x)/(1+\exp(x))$. Now we have $\psi''/\psi' = (1-\exp(x))/(1+\exp(x)) = (1-2\theta)$ by letting $\theta = \exp(x)/(1+\exp(x))$.

where $\theta = \psi(\omega(p,a)) \in [0,1]$ at $p = p_1$, and $a = 1$.

Therefore, $\partial\Pi^*(p,a)/\partial p < 0$ for $a \in (1, 1+\varepsilon)$ for small $\varepsilon > 0$ at $p = p_1$. Furthermore, by the assumption that $\Pi^*(p_1, 1) > 0$, $a^*(p_1) > 1$ and this is sufficienltly close to 1.[18] Then, as $\partial\Pi^*(p,a)/\partial a < 0$, $\partial a^*(p)/\partial p < 0$ at $p = p_1$.

By Theorem 4, $\partial\Lambda(\lambda, p)/\partial p > 0$. ■

[18] This is possible by taking K adequately.